

6G-Enabled Vehicle-to-Everything Communications: Current Research Trends and Open Challenges

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ABSTRACT With the developments on Advanced Driver-Assistance Systems (ADAS), autonomous driving and purely unmanned vehicles, such as the Unmanned Aerial Vehicles (UAVs), the pressure on the requirements for Vehicle-to-Everything (V2X) communications has drastically increased during the last few years. However, a new and appealing horizon is open for V2X applications with the advent of the Sixth Generation (6G) of mobile communications. In this paper, we present a review of V2X standards to offer a holistic perspective on the evolution of this technology toward 6G. Then, the key technological enablers for the 6G V2X are identified, namely (a) Non-Terrestrial Networks (NTNs), (b) Ultra Reliable Low Latency Communications (URLLC), (c) Artificial Intelligence (AI), (d) Integrated Sensing And Communications (ISAC) and (e) propagation channel modeling. For each of these technological enablers, the most recent proposals are thoroughly studied and the open challenges and opportunities identified. Our paper intends to serve as a timely roadmap for the development of future 6G V2X networks.

INDEX TERMS Artificial intelligence (AI), integrated sensing and communications, non-terrestrial network, sixth generation, ultra reliable low latency communications, vehicle-to-everything (V2X).

I. INTRODUCTION

Improving the safety and efficiency of vehicular transportation is a global priority, given the massive number of people and goods that are transported by road every day. For several decades, governments, industry, and academia around the world have joined forces to develop far-reaching solutions aimed at preventing traffic accidents and minimizing both economic losses and environmental impact caused by inefficient traffic management [1]. In this context, one of the most ambitious initiatives is to equip vehicles with communication capabilities that enable coordinated navigation through the

exchange of driving-relevant information [2]. The design of communication systems for connected and autonomous vehicles has been an active area of research and development for over 50 years. Significant progress has been made in this field during this time span by building upon the foundations of cellular mobile networks and wireless local area access technologies [3], [4]. In particular, Wi-Fi and Fourth Generation (4G) mobile cellular networks have propelled vehicular communications to a technology readiness level sufficient for the deployment of trial and pilot programs, e.g., see [5].

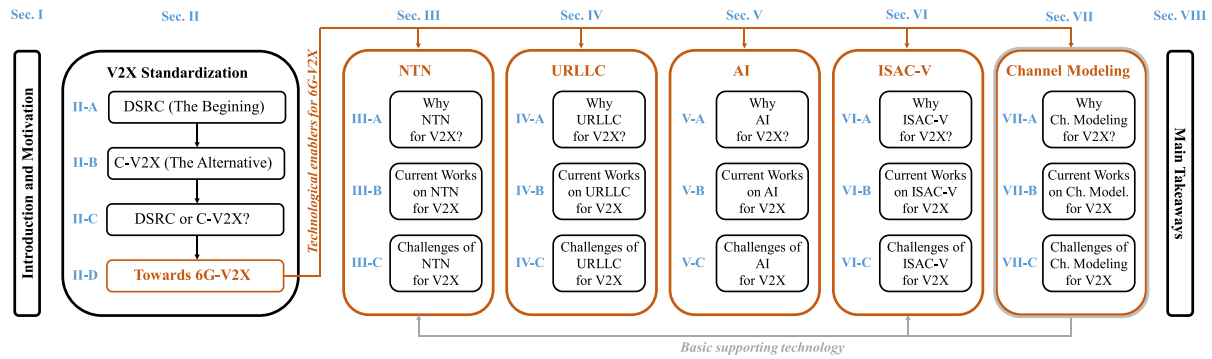


FIGURE 1. Structure and flow of contents of this review.

With the emergence of Fifth Generation (5G) cellular networks, the future of the V2X systems looks even more promising. However, due to their critical role in safety, V2X systems should be revised constantly to incorporate the latest technological breakthroughs and keep pace with the ever-changing demands of wireless service users. It is therefore important to consider current trends to devise how the V2X technology must adapt to incorporate the innovations brought by future generations of wireless communications. This is also important to take advantage of the opportunities created by tech enablers, such as artificial intelligence, sensing technologies, and aerial nodes for the expansion of wireless coverage. In particular, the advances on ADAS, autonomous driving and purely unmanned vehicles, such as UAVs, come along with the expectation of higher and higher performance, which exceeds the sky plot by the 5G. Novel V2X services claim for ubiquitous connectivity, ubiquitous intelligence, high-precision positioning and sensing, deterministic latency, robustness against complex and dynamic channel environments in V2X. The prospect of a new and appealing horizon for V2X applications is laid by the advent of a 6G of mobile networks [6]. 6G comes with the promise of supporting intelligent and endogenous secure connections for vehicles, full integration of communication and sensing/positioning, AI and Big Data-aided network slicing, and air-space-ground integrated architecture, among others.

STRUCTURE OF THIS PAPER, RELATED WORKS AND NOTATION

This paper provides an outlook on 6G V2X communications with the purpose of guiding the development of such communication systems. The structure and flow of contents of the paper is imaged in Fig. 1. First, Section II contextualizes the origin, trends and future directions of V2X by examining the underlying standards. These include the birth of Dedicated Short-Range Communications (DSRC), the evolution to the cellular-connected V2X and the early expectations for the 6G-V2X from the standardization point of view. The technological enablers introduced required to reach this 6G-V2X (marked in brown in Fig. 1) will then be sequentially analyzed. In particular, Section III will introduce the V2X integration with NTNs, Section IV URLLC for V2X, Section V the applications of AI for V2X, Section VI ISAC and ISAC for

Vehicles (ISAC-V), and Section VII the V2X channel modeling strategies. In the case of the V2X channel modeling strategies, they will constitute the basic technological support for the contents of some of the former sections (as imaged in grey in Fig. 1). For each technological enabler (Sections III–VII), we first introduce the basics, then thoroughly detail the latest developments in the area, to later end up by summarizing the main challenges and open opportunities. Finally, Section VIII provides a summary of the main conclusions of this paper. For convenience, the most relevant acronyms for the topic of the paper are specified in Table 1. Other acronyms are defined through the text.

Although other related surveys on V2X communications and related topics are available in the literature, such as [3], [4], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], this paper, instead of focusing on a single perspective, proposes a wider and more multidisciplinary analysis of the current status and future trends of V2X communications with respect to other works. Table 2 presents a comprehensive comparison between the contents of this work and other representative surveys in the literature at the time of writing.

II. CURRENT STATE AND FUTURE DIRECTIONS OF THE V2X COMMUNICATIONS STANDARDIZATION

The current state of the V2X communication systems and the path toward 6G can be better understood by examining the evolution of the underlying standards. This section provides a summary of the standardization milestones that have shaped the development of V2X communications. Several comprehensive survey papers have been published over the past 20 years, offering detailed descriptions of the different versions of the prominent V2X standards, e.g., see [3], [4], [7], [8], [9], [10], [11]. The aim of this section is to complement the information provided in such surveys by discussing recent developments in V2X technology and providing a holistic view of the V2X communications landscape on the path to 6G.

A. DEDICATED SHORT-RANGE COMMUNICATIONS STANDARDS FOR INTELLIGENT TRANSPORT: THE BEGINNING

Vehicular communications are closely intertwined with the concept of intelligent transportation systems (ITS). This concept was introduced in the 1980 s to describe “a group of

TABLE 1. Acronyms Definition

6G	Sixth Generation
A2G	Air-to-Ground
A2S	Air-to-Sea
ADAS	Advanced Driver-Assistance Systems
AI	Artificial Intelligence
AS	Active Sensing
BS	Base Station
CAV	Connected and Autonomous Vehicle
C-V2X	Cellular V2X
D2D	Device-to-Device
DPS	Direct Passive Sensing
DRL	Deep Reinforcement Learning
EES	Edge Enabler Server
eMBB	enhanced Mobile Broadband
F2M	Fixed-to-Mobile
FL	Federated Learning
HAPS	High-Altitude Platform Stations
HEO	Highly Elliptical Orbit
HRLLC	Hyper Reliable Low Latency Communication
IoT	Internet of Things
ISAC	Integrated Sensing And Communications
ISAC-V	ISAC for Vehicles
LEO	Low Earth Orbit
MEC	Mobile Edge Computing
MEO	Medium Earth Orbit
MIMO	Multiple-Input Multiple-Output
ML	Machine Learning
NGSO	Non-Geostationary Orbit
NOMA	Non-Orthogonal Multiple Access
NTN	Non-Terrestrial Network
OPS	Opportunistic Passive Sensing
OTFS	Orthogonal Time Frequency Space
PMN	Perceptive Mobile Network
QoS	Quality of Service
RAN	Radio Access Network
RIS	Reconfigurable Intelligent Surface
RSU	Road Side Unit
SAGIN	Space-Air-Ground Integrated Network
UAS	Unmanned Aircraft Systems
UAV	Unmanned Aerial Vehicle
UE	User Equipment
UM-MIMO	Ultra-Massive MIMO
URLLC	Ultra Reliable Low Latency Communications
USV	Unmanned Surface Vehicle
V2I	Vehicle-to-Infrastructure
V2N	Vehicle-to-Network
V2P	Vehicle-to-Pedestrian
V2V	Vehicle-to-Vehicle
V2X	Vehicle-to-Everything
VEC	Vehicular Edge Computing

technologies—including information technology, communications, control, and electronics—that connect vehicles to transportation infrastructure to enhance the safety and efficiency of transport systems” [20]. With the aim of promoting the development of ITS, in the fall of 1994, the U.S. National Intelligent Vehicle-Highway Systems Program was restructured into the Intelligent Transportation Systems Joint Program Office (ITS-JPO). One of the first tasks assigned to

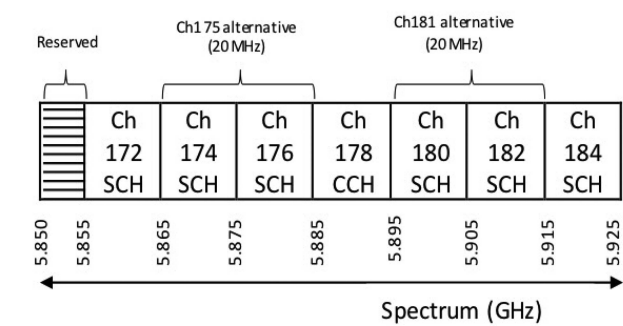


FIGURE 2. Frequency channels allocated in 1999 by the FCC for DSRC technology (SCH and CCH stand for service and control channel, respectively).

this new office was to promote the standardization of ITS. By the mid-1990 s, the U.S. ITS-JPO had developed and standardized a systems architecture to foster ITS interoperability [20]. This architecture includes three layers:

- A communications layer for the transmission of information between systems,
- A transportation layer responsible for defining subsystem and interface solutions for each transport service,
- An institutional layer, which encompasses institutions, regulations, funding mechanisms, and implementation procedures for ITS.

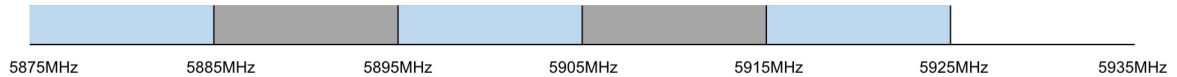
A major milestone in the development of the communications layer of ITS—and consequently, the creation of vehicular communication systems—was the request submitted in October 1997 by the Intelligent Transportation Society of America to the Federal Communications Commission (FCC) to allocate 75 MHz of spectrum in the 5.9 GHz band for DSRC intended for ITS applications [21]. This request was approved by the FCC in October 1999, establishing the use in the U.S. of the frequency band from 5.850 to 5.925 GHz for ITS radio services supported by DSRC technology [22]. The assigned spectrum was divided into seven 10 MHz channels plus a 5 MHz guard band. This partition, which remained effective until early 2025, is illustrated in Fig. 2 , along with the corresponding channel numbers designated by the FCC.

The allocation of spectrum to DSRC technology ignited the work for developing a standard for V2X communication. The first and most notable of these standards was published in 2010 as an amendment to the IEEE 802.11 standard for Wireless Access in Vehicular Environments (WAVE) [23]. This amendment, identified as IEEE 802.11p-2010, was incorporated into the main IEEE 802.11 standard in 2012 [24]. This standard is aimed at providing wireless access protocols for vehicular applications through specifications for the Physical Layer (PHY) and the Medium Access Control (MAC) layer of both Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communications in the 5.9 GHz band. The standard specifies the use of Orthogonal Frequency Division Multiplexing (OFDM) as the transmission scheme for V2V and V2I communications. With these specifications, V2V and V2I information exchange can occur at relative speeds of up to

TABLE 2. Comparison of This Survey With Other Representative Surveys on V2X Communications

Reference	Overview of V2X Standardization			Perspectives of 6G V2X	NTN for V2X	URLLC for V2X	AI for V2X	ISAC for V2X	V2X Channel Modeling
	IEEE 802.11	LTE	5G						
Kenney [3]	✓	×	×	×	×	×	×	×	×
MacHardy et al., [4]	✓	✓ (<Rel. 14)	×	×	×	×	×	×	×
Castañeda et al., [7]	×	✓	✓	×	×	×	×	×	✓
Soto et al. [8]	×	✓	✓	×	×	×	×	×	×
Hajisami et al. [9]	×	✓	×	×	×	×	×	×	×
Noor-A-Rahim et al. [10]	×	×	✓	✓	✓	×	✓	✓	×
Annu et al. [11]	×	✓	✓	✓	×	✓	✓ (briefly)	✓ (briefly)	×
A. M. Ince et al. [12]	×	×	×	✓	✓	✓	×	×	×
Sasiain et al. [13]	×	×	×	×	×	✓	×	×	×
Rafique et al. [14]	×	×	×	×	×	✓	×	×	×
Nasrallah et al. [15]	✓	✓	×	×	×	✓	×	×	×
Sedar et al. [16]	✓	✓	✓	×	×	×	✓	×	×
Hakeem et al. [17]	✓	✓	✓	✓	×	✓	✓	×	×
Feng et al. [18]	✓	×	✓	✓	×	✓	✓	×	×
G.-Guzman et al. [19]	×	×	✓	✓	×	×	✓	×	×
This survey	✓ (briefly)	✓ (briefly)	✓ (briefly)	✓	✓	✓	✓	✓	✓

For Road ITS:



For Urban Rail ITS:

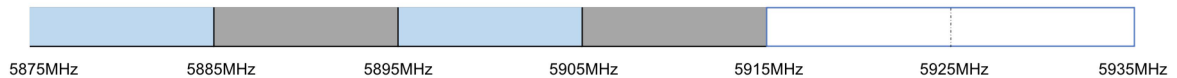


FIGURE 3. Frequency channels allocated by the ECC for ITS services.

500 km/h. The main aspects of the 802.11p standard have been reviewed in detail in several papers, e.g., [3].

In parallel with the regulatory and standardization efforts in the U.S., in March 2008, the ECC of the European Conference of Postal and Telecommunications Administrations (CEPT) allocated 30 MHz in the 5.9 GHz band for ITS use on a non-exclusive basis within the European Union, spanning the frequency band from 5.875 to 5.905 GHz [25]. Following this decision, in 2020, the European Commission extended the bandwidth allocation up to 5.935 GHz, giving priority to Road ITS applications in the 5.875–5.915 GHz sub-band and to Urban Rail ITS applications above 5.915 GHz [26]. The 5.925–5.935 GHz sub-band was reserved specifically for Urban Rail ITS applications. Fig. 3 shows the frequency channels currently designated in Europe for ITS use. Capitalizing on these spectrum harmonization directives of the

CEPT, the European Telecommunications Standards Institute (ETSI) established a V2X communications standard for ITS applications by building on the guidelines of the IEEE 802.11 standard. This European standard, known as ITS-G5 [27], is therefore similar to the DSRC technology developed in the U.S. ETSI's endeavors to standardize vehicular communication systems are guided by a vision aimed at providing three main types of ITS services, namely: road safety, traffic efficiency, and others [28].

The DSRC technology for ITS has also been supported and adopted in other regions of the world. For example, Japan has contributed to its development. In this country, the Association of Radio Industries and Business (ARIB) issued specifications for vehicular networks operating in the 700 MHz band [29]. Several DSRC trial networks built upon the IEEE 802.11p standard in the 5.9 GHz band have been

deployed. The Ann Arbor Safety Pilot Model Deployment in Michigan [30] and the New York City pilot [31] are some notable examples.

B. CELLULAR V2X STANDARDS FOR INTELLIGENT TRANSPORT: THE ALTERNATIVE

The evolution of V2X communications split into two distinct paths in the mid-2010s, when the Third Generation Partnership Project (3GPP) incorporated Device-to-Device (D2D) communication capabilities into its LTE standard for 4G mobile cellular networks. These capabilities were introduced in [32] through a new interface that allows one User Equipment (UE) to connect with another UE. This interface is referred to as *sidelink* in 3GPP terminology and is also known as the proximity communication (PC5) interface [28]. The use of this interface for LTE-based V2X communications was specified in a subsequent release, Release 14, that the 3GPP issued in 2018 [33]. This release defines two operational modes for vehicular communications. The first mode is based on the Uu interface defined by the LTE standard for connectivity with the public cellular network [34], [35]. The Uu mode is a communication interface between terminals and Base Stations (BSs), characterized by long-range and wide coverage reliable communication. The second mode is based on the PC5 interface to enable short-range V2X communications. The main difference between the PC5 and Uu modes can be summarized as follows:

- 1) Communication distance: In general, PC5 is used to enable short-range information exchange between terminals, providing a short-range direct communication among vehicles, pedestrians, and road infrastructure. By contrast, relying on BS and the core network, the Uu interface enables large-scale continuous coverage. Vehicles can maintain communication connections through BSs handover during mobility, addressing the distance limitation of PC5 communication.
- 2) Multimedia service capability: PC5 cannot interface directly with cloud services, and thus its multimedia service capability is limited. By contrast, as the Uu interface is able to connect to Mobile Edge Computing (MEC) and clouds, it supports real-time interaction between MEC and cloud services. Hence, the Uu-based networking can provide much richer V2X applications, such as edge-aided vehicle monitoring, traffic broadcasting, cloud-based decision control.
- 3) Cost and efficiency: As the Uu interface can reuse the well-deployed cellular network, it reduces the cost of V2X networking. By comparison, PC5 requires separate deployment of private network infrastructure, such as road side units (RSUs), leading to higher costs in wide-area coverage scenarios.

V2X communication systems based on cellular mobile communication technology are commonly referred to as Cellular V2X (C-V2X) systems. The original vision of the 3GPP for C-V2X technology based on LTE considered three types of vehicular communications:

- V2V: LTE-based communication between UEs that are exchanging V2V application information,
- V2I: LTE-based communication between a UE and a RSU,
- Vehicle-to-Pedestrian (V2P): LTE-based communication between a vehicle and a portable LTE device carried by an individual, e.g., a pedestrian or a cyclist.

However, an additional type was later included in [36] as follows:

- Vehicle-to-Network (V2N): communication between an LTE UE and an application server that supports V2N applications.

The LTE-based C-V2X access layer utilizes Single Carrier Frequency Division Multiple Access (SC-FDMA) as its transmission scheme. In this framework, resource blocks (RBs) are defined to include 12 subcarriers each. The system supports scalability through carrier aggregation, which enables the addition of RBs to the symbol depending on the bandwidth. In a standard 10 MHz bandwidth configuration, the specifications allocate 50 RBs, resulting in a total of 600 subcarriers.

The development of C-V2X systems gained further momentum in 2019 with the publication by 3GPP of a set of specifications for 5G cellular mobile communications [37]. These specifications, also known as 5G New Radio (5G-NR) C-V2X [7], introduce advanced V2X communications that provide Quality of Service (QoS) support, enabling a V2X application to receive timely notifications of changes in QoS. In addition, the 5G-NR C-V2X specifications include more stringent performance requirements than those defined for LTE-based C-V2X, with the aim of delivering services beyond the capabilities of LTE-V2X, such as those targeting at autonomous vehicle applications. However, an important aspect of C-V2X systems based on LTE and 5G technologies is that they are designed to coexist seamlessly with one another. The 5G-NR C-V2X considers the stack of autonomy levels established by the Society of Automotive Engineers (SAE) in [38], namely: Level 0, No automation; Level 1, Driver Assistance; Level 2, Partial automation; Level 3, Conditional automation; Level 4, High automation; Level 5, Full automation. The 5G-NR C-V2X system seeks to provide support for all levels of automation. Two comprehensive surveys on the characteristics of the 5G-NR C-V2X system are provided in [7], [8].

C. DSRC OR C-V2X: THE RACE FOR V2X DOMINANCE

The growing interest in C-V2X communications contrasts with the recent decline in support for DSRC technology. This is due in part to various independent studies suggesting that C-V2X systems perform better than DSRC systems based on the IEEE 802.11p standard, e.g., see [39], [40]. In addition, the electromagnetic spectrum scarcity together with the growing demand for wireless communication services sparked a dispute in the U.S. over the reallocation of the 5.9 GHz band for use by next-generation Wi-Fi networks and C-V2X systems. The FCC settled the dispute in 2021 by designating the lower 45 MHz of the 5.850–5.925 GHz band (guard channel from

5.850-5.855 GHz and channels 172, 174, 176, and 178 in Fig. 2) for unlicensed use and reserving the upper 30 MHz (channels 180, 182 and 184) for ITS services, mandating as well a transition from DSRC to C-V2X technology [41].

The phase-out of DSRC technology operations in the U.S. was ratified by the FCC in November 2024 through a Second Report and Order, which was published in the Federal Register on December, 2024 [42]. In this Second Report and Order, the FCC determined that DSRC technology may no longer be used for ITS applications in the 5.9 GHz band after December 14, 2026. Moreover, it was ruled that the 5.895 to 5.925 GHz sub-band has been reserved for ITS services based on C-V2X technology effective February 11, 2025. Regarding the original channel designations within the 5.9 GHz band, the FCC maintains a 10 MHz channel bandwidth structure, resulting in three 10 MHz channels for ITS services based on C-V2X: 5.895-5.905 GHz, 5.905-5.915 GHz, and 5.915-5.925 GHz. These channels can be combined into contiguous 20 MHz channels or a single 30 MHz channel. The Commission also opted to discontinue the consideration of specific channel number designations.

While the shutdown of the DSRC technology is imminent in the U.S., the situation is different in other parts of the world. In Europe, for example, this technology remains active. In fact, Release 2 of the ITS-G5 standard has recently been approved by the ETSI [43]. This release incorporates new specifications of the IEEE 802.11 standard as defined in the IEEE 802.11bd-2022 amendment for enhanced V2X systems [44]. This improved version of the ITS-G5 standard gives a new air to DSRC. In fact, some studies show that the access layer of the IEEE 802.11bd standard outperforms C-V2X standards in terms of packet error rate and packet reception ratio [45].

D. TOWARD 6G-V2X STANDARDIZATION

The ITS-G5, LTE-C-V2X and 5G-NR standards are promising technologies for unleashing the potential of connected vehicles. However, they will hardly meet the ultra-low latency and extreme throughput requirements needed for intelligent and fully autonomous driving. Furthermore, these standards were primarily conceived for two-dimensional terrestrial communications. Nevertheless, the rapid growth of the aerospace and aeronautics sectors calls for three-dimensional V2X communications supporting UAV, nano-satellite, and high-altitude platform stations (HAPS) applications. These new requirements are fostering the development of a new generation of V2X communication standards that significantly enhance capabilities to deliver ultra-low latency, intelligent, reliable, and resilient communications. Such V2X systems should offer support for high-density deployments and ubiquitous coverage to ensure service availability anytime and anywhere. The path toward such advanced V2X systems is beginning to take shape through the ongoing standardization of the International Mobile Telecommunications networks for 2030 and beyond (IMT-2030), also known as 6G networks.

The International Telecommunications Union (ITU) established the framework and objectives for 6G network development in Recommendation ITU-R M.2160-0 [46]. This document sets forth a series of objectives and key performance indicators (KPIs) aimed at improving the capabilities of 5G systems pertaining to security and resilience, reliability ($1 \cdot 10^{-5}$ - $1 \cdot 10^{-7}$), latency (0.1 - 1 ms), mobility (500 - 1,000 km/h), connection density (10^6 - 10^8 devices/km²), area traffic capacity, spectral efficiency, user-experienced data rates, and peak data rates. Additionally, it defines new targets related to enhanced coverage, sensing-related capabilities, artificial intelligence integration, sustainability, interoperability, and positioning accuracy (1 - 10 cm). These capacities will be pivotal to realizing assisted intelligent and automated driving applications. It can be anticipated that the development of a V2X system based on 6G will be part of a second phase in the broader 6G standardization process. However, further research advancements are needed to expand knowledge and increase technology capacities across multiple dimensions not addressed in current V2X standards. Some of these needs have already been discussed in other works, such as [10], [11]. Other areas that present significant open research challenges, such as V2X for NTN, URLLC V2X, V2X AI and ISAC-V are explored in the following sections.

III. NON-TERRESTRIAL NETWORKS

A. NTNS AS A 6G V2X TECHNOLOGICAL ENABLER

NTNs are communication systems established through non-terrestrial nodes, aiming to overcome the limitations of terrestrial infrastructures and achieve seamless, global, three-dimensional coverage [47], [48]. They address the connectivity challenges in remote areas—such as oceans, deserts, and mountainous regions—as well as in special-use scenarios, including flying vehicles, ocean-going vessels, and disaster recovery operations. NTNs are instrumental in providing fundamental connectivity to nearly half of the global population that remains unconnected.

The deployment of NTNs relies on two primary platforms: spaceborne and airborne systems, which together form the three-dimensional architecture envisioned for 6G networks [49]. The specific types of NTNs platforms can be found in Table 3 and Fig. 4. The spaceborne segment provides large-scale global coverage via low earth orbit (LEO), medium earth orbit (MEO), geostationary earth orbit (GEO), and highly elliptical orbit (HEO) satellites, with a primary focus on bridging coverage gaps over oceans and sparsely populated terrains. The airborne segment leverages HAPS and unmanned aircraft systems (UAS) to reduce latency and support delay-sensitive applications such as flying vehicles and emergency response.

The 3GPP initiated NTN standardization in releases 15 and 16, which laid the foundation by identifying key technical challenges, including satellite propagation delay and Doppler shift [50]. Release 17 formalized the use of transparent payload architectures, introducing essential PHY, MAC,

TABLE 3. Characteristics of Different NTN Platforms

Platform	Altitude Range (km)	Orbit/Positioning	Key Features
HEO Satellite	Perigee > 500, Apogee < 50,000	Highly Elliptical Orbit	Communication in high-latitude regions
GEO Satellite	35,786	Geostationary	Wide coverage, fixed relative to Earth
MEO Satellite	8,000-20,000	Circular Orbit	Medium latency, navigation services
LEO Satellite	500-2,000	Circular Orbit	Low latency, suitable for real-time applications
HAPS	20	Stationary in stratosphere	Lower latency, wide regional coverage
UAS	0-1	Regional mobility	Flexible deployment, localized enhancement

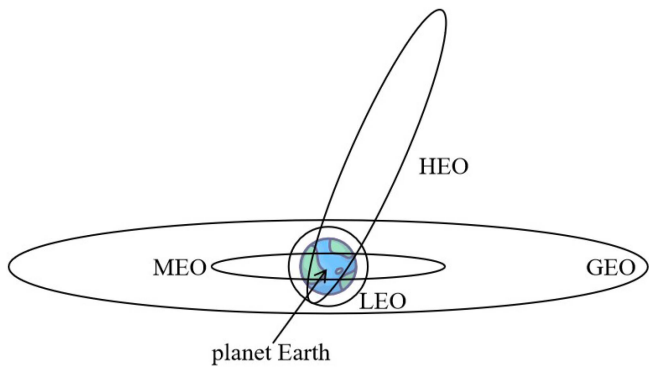


FIGURE 4. Illustration of the classes of satellites.

and Radio Resource Control (RRC) enhancements, such as timing advance adjustments, extended random-access channel (RACH) windows, and System Information Block # 19 (SIB19) for ephemeris data, to address GEO and LEO-specific latency constraints. Release 18 expanded NTN support through uplink coverage enhancements for mobile devices and cross-network mobility features. The upcoming release 19 introduces regenerative payloads featuring onboard gNodeB (gNB) processing, enabling lower latency and compatibility with other devices. Looking forward, release 20 is bifurcated into two parts: part 1 advances 5G-Advanced with multi-orbit coordination and support for mission-critical services, while part 2 lays the foundation for 6G by defining native NTN-terrestrial integration frameworks and AI-enhanced Radio Access Network (RAN) requirements [51]. These efforts converge in release 21, which will formally establish NTN as an integral component of the 6G architecture, enabling global URLLC under 10 milliseconds.

B. PROPOSALS AND RECENT ADVANCES ON NTNS FOR V2X

The convergence of reconfigurable intelligent surface (RIS), advanced Multiple-Input Multiple-Output (MIMO) architectures, and next-generation multiple access (NGMA) technologies represents a critical research frontier for 6G NTNs [52]. This integration is driven by the need to overcome fundamental limitations inherent in space-air-ground communications. Conventional NTNs are challenged by severe path loss—particularly at millimeter-wave (mmWave) and Terahertz (THz) frequencies—spectrum scarcity, and mobility-induced

channel instability, all further exacerbated by the limited power budgets of orbital and aerial platforms. RIS technology, encompassing both passive and active variants, has emerged as a promising solution for intelligent electromagnetic wave manipulation, enabling dynamic beam redirection around obstacles and low-power signal amplification [53], [54]. Simultaneously, ultra-massive MIMO (UM-MIMO) systems exploit extreme spatial diversity to achieve significant spectral efficiency gains, while NGMA techniques such as non-orthogonal multiple access (NOMA), rate-splitting multiple access (RSMA), and space division multiple access (SDMA) enhance user connectivity through non-orthogonal resource allocation [55], [56], [57].

The handover (HO) strategies for Non-Geostationary Orbit (NGSO) satellite systems pose a critical challenge in the standardization of 6G NTNs, particularly in the context of emerging mega-constellations [58]. Legacy HO techniques—originally designed for small-scale satellite systems such as Globalstar (fewer than 100 satellites)—are inadequate for managing the dynamics of large-scale LEO, MEO, and HEO constellations. These modern systems feature thousands of satellites, multi-orbit topologies, and diverse regulatory environments. Prior studies [59], [60], [61] have identified major pain points, including frequent handovers caused by satellite mobility (typically every 2–4 minutes in LEO), propagation delays ranging from 25 to 250 ms, and significant control signaling overhead due to simultaneous user handovers across expansive coverage footprints. Although 3GPP Releases 16 and 17 introduced initial HO trigger mechanisms (e.g., location- and timing-based criteria), they lacked comprehensive satellite selection strategies and failed to quantitatively evaluate packet-level KPIs such as spectral efficiency, Doppler stability, service interruption time, and HO frequency. More recent solutions [62], [63] incorporating Machine Learning (ML) and software-defined networking (SDN) remain constrained to outdated constellations or small-scale simulations (fewer than 200 satellites), leaving substantial gaps in the understanding of HO performance under real-world NGSO configurations.

The application of AI in 6G NTNs plays a pivotal role in addressing the operational complexities introduced by orbital dynamics, constellation-scale interference, and heterogeneous service demands [64]. Standardized in 3GPP release 18, AI-driven NTN architectures enable autonomous network orchestration via deep learning-based predictive resource allocation,

facilitating adaptive spectrum sharing across LEO satellites while alleviating congestion. Federated learning frameworks further leverage orbital regularities, allowing satellites to aggregate localized models via optical ground stations and collaboratively generate real-time interference maps—thereby eliminating the need for centralized data processing. Reinforcement learning enhances RAN intelligence by optimizing beamforming and handover strategies in response to Doppler shifts and signal attenuation, achieving up to a 90 % reduction in handover failures. Moreover, AI-native cross-layer interfaces empower UE and gNB co-inference for compressing Channel State Information (CSI) feedback (reducing overhead by 50 %) and enabling precise localization [65]. Graph neural networks (GNNs) proactively detect link anomalies up to 15 seconds before service disruption, enhancing network reliability [66].

The integration of NTN with V2X communications addresses critical limitations in terrestrial vehicular systems, particularly for autonomous driving applications that require ubiquitous URLLC [12]. Terrestrial V2X often struggles with Non-Line-of-Sight (NLoS) obstructions in urban environments and total connectivity blackouts in rural or remote areas (e.g., highways, deserts, oceans), and faces scalability constraints during large-scale vehicular coordination scenarios. NTN platforms—particularly HAPS at approximately 20 km altitude and LEO satellites—offer three-dimensional coverage that eliminates such connectivity gaps. HAPS, with their stratospheric persistence and edge computing capabilities, enable sub-10 ms latency, which is crucial for time-critical applications such as collision avoidance. LEO constellations complement this by ensuring global service continuity. This convergence enables a wide range of advanced use cases, including precision agriculture using HAPS-controlled autonomous farming vehicles, cross-border logistics monitoring, and disaster-resilient communication infrastructure where NTN serves as a backup for terrestrial failures.

C. OPEN CHALLENGES AND OPPORTUNITIES ON NTNS FOR V2X

The integration of NTNs—exemplified by HAPS—with V2X communications presents a range of fundamental challenges that threaten both interoperability and large-scale deployment feasibility. Foremost among these is the conflict between technical compatibility and system synchronization. HAPS operates over wide-area, high-altitude stratospheric coverage, inherently differing in scale, dynamics, and latency requirements from V2X's low-altitude, real-time, ground-level interactions. This disparity complicates protocol alignment, timing coordination, and data synchronization across the integrated system.

Additionally, the convergence faces dual-layer security architecture constraints. HAPS systems require robust long-distance communication with high anti-jamming and fault tolerance capabilities, whereas V2X necessitates low-latency, secure direct connections capable of resisting localized threats such as spoofing and Sybil attacks. These divergent security

paradigms call for the development of unified, cross-domain security protocols that can operate effectively across both domains without compromising integrity or performance.

Finally, integration is hindered by significant cost and standardization bottlenecks. HAPS platforms entail substantial upfront investment and operational costs, while V2X systems are characterized by fragmented and heterogeneous deployment models. Aligning these two cost structures requires coordinated economic optimization. Simultaneously, the lack of cross-industry standards—spanning aviation and transportation sectors—further complicates spectrum allocation, data interface compatibility, and regulatory compliance. Without harmonized frameworks, the functional integrity and scalability of NTN–V2X integration remain at risk.

IV. ULTRA RELIABLE LOW LATENCY-ORIENTED V2X COMMUNICATIONS

A. URLLC AS A 6G V2X TECHNOLOGICAL ENABLER

URLLC is one of the core application scenarios of V2X [7]. Taking autonomous driving as an example, it requires a reliability of 99.999 % for transmitting 32-byte data packets, and the user plane latency is less than or equal to 1 millisecond (ms). The overall end-to-end latency mainly includes access latency, signal propagation latency, transceiver signal processing latency, queuing latency, retransmission latency, etc [67]. To ensure the realization of URLLC design objectives, substantial efforts have been invested, including but not limited to finite block-length (FBL) transmission, frame structure optimization, end-to-end resource slicing, and low latency multiuser access.

1) FINITE BLOCKLENGTH TRANSMISSION IN V2X

Classical information theory and communication systems are designed for large packet transmission. Given sufficiently large blocklength, an almost deterministic and error-free data transmission can be achieved [68]. To further reinforce reliability, classical systems often pair strong channel coding with retransmission strategies to retransmit corrupted segments. However, large blocklength inherently introduces processing delays, existing both in encoding at the transmitter to generate large codewords, and in decoding at the receiver to perform error correction. Retransmissions further exacerbate this issue, as each round-trip for feedback and retransmission adds tens-hundreds ms levels of latency, making such systems ill-suited for latency-sensitive applications in V2X. In a nutshell, reliable communication is guaranteed with larger blocklength, strong channel coding and retransmission strategy, which however sacrifice transmission latency.

Also, in classical large blocklength communications, the overhead of metadata (e.g., pilot and control information) is negligible compared to the information payload. For example, a 10,000-bit data packet paired with 100 bits of metadata incurs only a 1% overhead. This tradeoff can be easily justified by the requirement for reliable channel estimation

and accurate synchronization. In contrast, FBL communication upends this balance, as overhead becomes prohibitively prominent [69]. Considering a typical sensor data transmission case, a 100-bit payload may be accompanied by the same 100 bits of metadata, where the overhead soars to 100%, with doubled transmission time.

In V2X scenarios, there is a significant demand for structured data transmission. These structured data are characterized by small payloads, frequent interactions, and requirements for high reliability and low latency. As a result, classical Shannon theories and technologies that are designed for asymptotically large blocklengths become inapplicable, which motivated the research of FBL information theory. The limits of communication for a coding block of size n was identified in [70], which is a function with respect to block length, block error probability (BLEP) and Signal-to-Noise Ratio (SNR) over additive white Gaussian noise (AWGN) channel, written as

$$R_{\text{NA}} = C - \sqrt{\frac{V}{n}} Q^{-1}(P_e) + \frac{\log_2(n)}{2n} + o\left(\frac{1}{\sqrt{n}}\right), \quad (1)$$

where R denotes the rate, C denotes the channel capacity, V denotes the channel dispersion coefficient, and P_e denotes the average error probability. The rate demonstrated in (1) is known as the normal approximation (NA) bound [71], which is an asymptotic approximation for the achievability by a suitable encoding and decoding couple. As shown in (1), FBL brings low latency, at the cost of reduced reliability. Hence, existing research has focused on several research directions, such as FBL coder/decoder, frame structure optimization, and cross-layer design for reliability enhancement, to strike the latency-reliability trade-off.

2) RESOURCE SLICING AND MANAGEMENT IN V2X

V2X involves diverse multimedia services with varying requirements for latency, bandwidth, and reliability, where end-to-end resource slicing is a key enabler for ensuring QoS requirement of heterogeneous services [35]. In V2X networks, end-to-end resource slicing is mainly composed of radio access network, bearer network, and core network. A straightforward interpretation of slicing is the visualizing networks into end-to-end logical network slices, through differentiated configurations of the air network, bearer network, and core network, which represents an evolution from resource blocks slicing of radio network.

In general, the majority of V2X applications can be mapped to typical 5G scenarios. For examples, enhanced mobile broadband (eMBB) slicing is suitable for the high-definition maps and video services; massive machine-type communication (mMTC) slicing is suitable for the information interaction of massive vehicle sensors; URLLC slicing is suitable for remote driving information services [72]. However, ensuring guaranteed QoS and achieving service-level agreement (SLA) performance of V2X remain non-trivial challenges. For the vehicle-to-BS radio link, it includes time, frequency,

space, and power resources [73], [74] [75]. For the bearer network, it includes optical transmission resource, routing and switching resource, and virtual link resources, etc. In the core network, it mainly includes inter-network element interface resources, such as switching and user-plane function (UPF) resources. In rare cases, end-to-end resource management can be formulated as a convex problem, which is readily solvable by existing algorithms [76], [77], [78]. In general, end-to-end resource slicing management is typically modeled as a complicated non-convex optimization problem, addressed by semi-dynamic resource reservation [79], or by learning-based techniques, such as reinforcement learning [80] and Q-learning [81]. In addition to the above resource slicing designs, a significant amount of research has focused on multiuser access, which is a key resource in the radio access network. In fact, due to the randomness of wireless links, the latency and reliability of end-to-end services are predominantly determined by the wireless link. Hence, the existing work includes dynamic multiuser access scheduling [75], semi-persistent scheduling (SPS), non-orthogonal multiuser access [82], grant-free access [83], etc. These techniques address the conflicting demands of low latency, high reliability, and massive connectivity in diverse scenarios.

B. PROPOSALS AND RECENT ADVANCES ON URLLC FOR V2X

In this section, we outline the recent progress on the topic of URLLC design of V2X.

1) FINITE BLOCKLENGTH CODER AND DECODER

With sufficiently large blocklength, modern channel coding techniques, such as low density parity check code (LDPC) codes and Turbo codes, can drive error rates to almost-zero levels, enabling nearly deterministic, error-free data delivery [68]. In fact, this reliability comes from the ability of long blocks to average out channel impairments, where random noise, fading, and interference are statistically smoothed over sufficiently long transmission durations, allowing coding schemes to exploit large degrees of freedom for error correction.

For well balancing the latency and reliability performance, it is necessary to research FBL coding and decoding designs. FBL codes with strong error-correction capabilities are essential in URLLC applications of V2X to meet the stringent latency and reliability requirements [84]. Though LDPC codes have been adopted for coding scheme in 5G systems, it has a gap larger than 1 dB to the bound at FBL regimes. Considering the fact that classical LDPC and turbo codes show a certain performance gap with respect to the NA bound, polar codes, Bose-Chaudhuri-Hocquenghem (BCH) codes, cyclic redundancy check (CRC)-aided polar codes and Polarization-adjusted convolutional (PAC) codes are considered as promising candidates for FBL [85] regimes, with the ability to approach the NA bound. As demonstrated in [84], polar codes with successive cancellation list (SCL) decoding

provide a reliability of $(1 - 10^{-4})$ with only 0.5dB gap to the NA bound, at a reasonable level of complexity.

On the other hand, FBL also requires novel decoders. A desired decoder for FBL is to achieve the near maximum-likelihood performance of the codes, while keeping a low level of decoding complexity. The popular decoders for FBL regimes include but are not limited to ordered statistics decoder (OSD), guess random additive noise decoding (GRAND), SCL decoding, successive cancellation stack (SCS) decoding, and sequential decoding (SQ) decoding. In general, the BLEP performance varies with different code lengths. Taking (64, 36) codes as examples, BLEP performance of SCL, SCS, and SQ are almost identical, approaching the maximum likelihood of BCH code [85]. Also, when comparing the performance of different decoding, it is necessary to consider complexity. The number of operations include the total number of additions, multiplications, and comparison in decoding data. The complexity of SCL, SCS, and SQ heavily rely on the code structure.

2) FBL FRAME STRUCTURE DESIGN FOR V2X

In the FBL regimes, blocklength directly determines the data preparation time, i.e., shortening blocklength effectively reduces packet preparation time, and makes the user access more compatible to low latency multiuser access. However, it reduces latency at the cost of degraded reliability performance, as demonstrated by the NA bound (3). Also, while the size of overhead is comparable to the data size in FBL scenario, the quality of channel estimation directly dictates the reliability of signal decoding. This makes pilot overhead a non-negligible component of FBL design, as it directly trades off with data throughput while anchoring the reliability of the entire transmission process [86].

In general, to achieve low latency and high reliability, the frame structure adheres to three critical design principles. The first is the self-contained transmission principle: Data within a slot or beam can be decoded independently without relying on other slots or beams. This means that a given slot and beam already contain all reference signals (e.g., demodulation reference signal) required for data demodulation, eliminating cross-dependency between resources. The second is the time-frequency concentration principle: Control channels are localized in the time-frequency domain rather than mapped across the entire system bandwidth. This design facilitates the introduction of new transmission types while maintaining backward compatibility with existing ones, ensuring flexible coexistence of diverse services. The third is the flexible timing relationship principle: Static or rigid timing constraints between slots and transmission directions are avoided. For example, asynchronous hybrid automatic repeat request (HARQ) replaces LTE's synchronous HARQ, which required pre-defined retransmission timings. This flexibility enables dynamic adaptation to traffic patterns and reduces latency by eliminating fixed retransmission cycles.

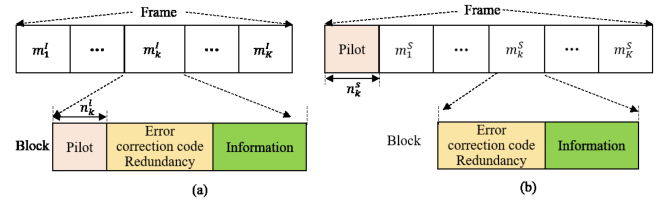


FIGURE 5. Illustration of shared pilot and independent pilot structures.

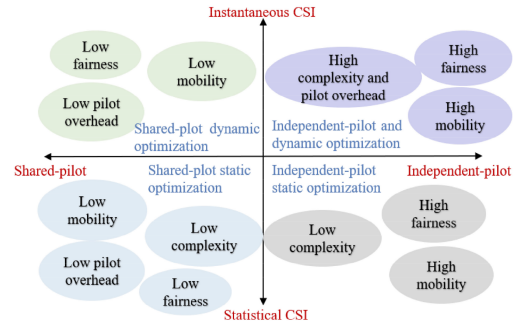


FIGURE 6. Shared-pilot and independent-pilot structures feature different applicability scenarios.

Due to the short blocklength in FBL regime, channel coefficients across adjacent blocks may exhibit temporal correlation within the coherence time, which introduces both challenges and opportunities for pilot design. [87] proposed to optimize the training sequence length and the rate adaptation scheme with respect to the delay performance. As shown in Fig. 5, for independent pilot structures, a dedicated pilot is embedded at the beginning of each block to enable per-block channel estimation, which is suitable for rapidly time-varying channels [87]. By contrast, shared pilot structures allocate a common pilot at the beginning of a frame, and this pilot is shared across all K blocks within the frame, leveraging channel coherence to reduce pilot overhead in quasi-static or slow-fading channels [88].

Recent research has formulated the effective throughput as a function of blocklength n and pilot length [89]. This framework enables optimization of effective throughput under latency (relates to blocklength) and reliability constraints (relates to BLEP). Solving this optimization problem fundamentally resolves the core questions: Under which scenarios are the independent-pilot and shared-pilot frame structures respectively optimal in FBL regime. The effective throughput performance is demonstrated in Fig. 7, which yields the following observations.

Low-mobility vs. high-mobility: in general, in low-mobility environments, where channel coherence time is substantially prolonged, the shared pilot structure is preferable. This approach allows all blocks within a frame to leverage a shared pilot, enabling decoding with minimal BLEP degradation. By contrast, high-mobility scenarios shorten channel coherence time, causing CSI to vary significantly across blocks. Here, the

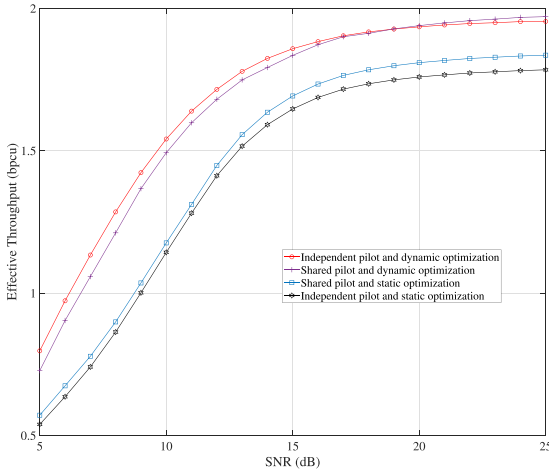


FIGURE 7. Effective throughput performance by different shared-pilot and independent-pilot frame structures. Doppler frequency $f_d = 30$ Hz.

independent pilot structure tends to be optimal, as per-block pilot insertion maintains BLEP within acceptable bounds.

Overhead and fairness requirement: the shared-pilot design offers inherent advantages under strict pilot overhead constraints, whereas independent pilots are favored when overhead limitations are relaxed. Regarding fairness performance, in the shared pilot structure, blocks proximal to the pilot exhibit superior decoding accuracy due to fresher CSI, and users assigned to these front blocks obtain better BLEP performance. In contrast, blocks positioned later in the frame suffer from accumulated channel estimation errors, and thus users assigned to these rear blocks obtain inferior BLEP performance.

Dynamic optimization vs. static optimization: when conducting the frame structure optimization, complexity is an essential and practical metrics. Dynamic optimization scheme leverages instantaneous CSI, and enables superior performance, but entails high computational complexity and overhead. Conversely, offline optimization relies solely on statistical CSI characteristics, requiring re-optimization only when underlying CSI statistics change. This approach ensures low complexity and minimal signaling overhead, albeit at the cost of performance degradation relative to dynamic counterparts.

3) URLLC-ORIENTED USER ACCESS

The multiuser access latency accounts for more than 30 % of the total end-to-end latency. In classical dynamic scheduling mechanisms, the system allocates wireless resources to users via uplink and downlink resource grants (RGs) in each transmit time interval (TTI), leading to redundant signaling overhead and latency. As early as in the 3GPP R8 protocol, SPS has been proposed for voice services (VoIP) to minimize signaling overhead in LTE systems [90]. To be specific, SPS periodically assigns fixed resource blocks (RBs) to specific users, featuring “one-time allocation, multiple-time usage.”

Although this sacrifices some flexibility, it significantly reduces access latency and control signaling overhead [91].

a) Latency performance of SPS and proactive scheduling for V2X: In fact, periodic communication services are prevalent in V2X application scenarios, including real-time broadcasting (e.g., speed, position, heading), periodic traffic information release by RSUs (e.g., traffic light status, congestion warnings), cloud information push, and data interaction in platoon cooperative driving [72]. These services require high-frequency data exchange, low-latency, and stable communication, for which the SPS mechanism is particularly suitable to enhance transmission efficiency and reliability. However, when a user initiates uplink data transmission in the first time, it needs to wait for the scheduling request (SR) slot, which are transmitted periodically (typically set to 20–80 ms in 3GPP specifications, depending on the number of users or service requirements). To mitigate this waiting latency, proactive scheduling mechanisms can be further introduced to enable users to transmit data directly without the needs of waiting SR.

Table 4 presents typical values of packet scheduling latency for different multiuser access mechanisms. In dynamic scheduling-based multiuser access, the latency for transmitting N_{blocks} data segments is approximated calculated as $(T_{\text{sr}}/2 + K_2) \times N_{\text{blocks}}$, where T_{sr} denotes the SR configuration period (the periodicity at which SR is configured). K_2 is the slot offset value representing the time interval from when a user receives a resource grant to when it actually starts transmitting uplink data. N_{block} denotes the number of radio link control (RLC) segments of packets. When SPS scheme is enabled, the user needs average waiting time $T_{\text{sr}}/2$ for initializing its transmission, and then the subsequent transmission time is calculated as $T_{\text{SPS}} \times \lceil N_{\text{slots}}/N_{\text{blocks}} \rceil$. T_{SPS} denotes the SPS period, i.e., the period length for the repetitive allocation of uplink resources by the BS to the user. N_{slot} denotes the number of uplink slots per SPS period.

With SPS and proactive scheduling scheme enabled, the user can directly transmit data packets in the first uplink slot after data preparation is completed. The user-plane latency is significantly reduced, written as $T_{\text{SPS}}/2 + T_{\text{SPS}} \times \lceil (N_{\text{block}} - 1)/N_{\text{slots}} \rceil$.

Note that the latency reduction of SPS and pre-scheduling entails trade-off. That is, if a user has no data to send, it still needs to transmit padding data to maintain resource occupation, leading to additional uplink interference and necessary power consumption. Hence, Release 16 further introduces dynamic grant skipping mechanism, allowing users without data to dynamically skip slots during the configured grant period. Meanwhile, an uplink extension timer is introduced. When users temporarily stop transmitting data, the timer maintains resource allocation to avoid frequent connection reestablishment, balancing latency and energy efficiency, as illustrated in Fig. 8. Low-latency performance is achieved while the extension timer is running, and user battery savings are realized after the timer expires.

TABLE 4. Average Latency for Multiuser Access by Different Schemes

Schemes	Average latency	Advantages	Disadvantages
Dynamic scheduling	$(T_{SR}/2 + K_2) \times N_{blocks}$	High resource utilization and flexibility	High multiuser access latency
SPS	$T_{SR}/2 + T_{SPS} \times \lceil N_{slots}/N_{blocks} \rceil$	One-time allocation, multiple-time usage	High latency when initializing first segment transmission
SPS+proactive scheduling	$T_{SPS}/2 + T_{SPS} \times \lceil (N_{block} - 1)/N_{slots} \rceil$	Direct transmission without the need of waiting SR	Dynamic grant skip is needed for reducing unnecessary power consumption and interference.

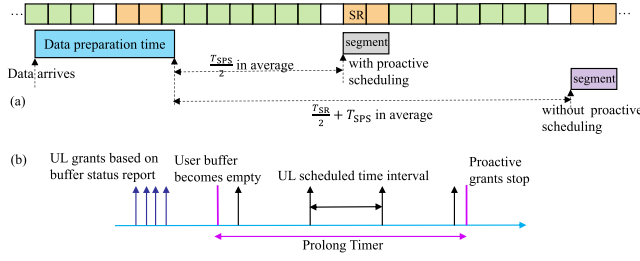


FIGURE 8. (a) The illustration of SPS and proactive scheduling, where the access more than 10 ms latency can be reduced depending on the systems configuration. (b) Dynamic grant skipping mechanism and uplink extension timer can be introduced, to reduce the interference and power consumption.

b) Reliability performance of SPS and proactive scheduling: When SPS is enabled for uplink transmission, once a user initiates a service, data can be transmitted on the allocated resource blocks. The packet transmission reliability is characterized by

$$P_{UL} = (1 - P_1) + P_1(1 - \epsilon)p_{12}, \quad (2)$$

where P_1 denotes the probability of initial transmission failure, ϵ is the probability of RG reception error when the base station reallocates resources for retransmission after the first failure, and p_{12} is the retransmission success probability. Evidently, uplink reliability depends on initial packet reliability, RG reliability, and data reliability after HARQ retransmission. For downlink transmission, users monitor RB status within a scheduled time. Once the BS initiates transmission, data is sent immediately on the RBs. If the initial transmission fails, the user feedbacks a Negative ACKnowledgement (NACK) to request retransmission, yielding the reliability as

$$P_{DL} = (1 - P_1) + P_1(1 - \epsilon_{NACK})(1 - \epsilon)p_{12}, \quad (3)$$

where ϵ_{NACK} is the probability of NACK feedback failure, and ϵ is the probability that users fail to correctly detect or parse new RBs when the base station reallocates resources for retransmission after receiving a NACK. These expressions show that SPS effectively improves URLLC packet transmission reliability, though at the cost of flexibility and spectral efficiency to some extent. Also, when the proactive scheduling is further enabled, the reliability performance is similarly calculated.

C. OPEN CHALLENGES AND OPPORTUNITIES ON URLLC FOR V2X

While extensive research has been paid to advancing URLLC for V2X, a critical gap remains in the holistic integration of in-vehicle networks and Uu air interface. This gap reflects both an engineering challenge, rooted in the need to synchronize heterogeneous hardware and protocol designs, and a theoretical challenge, stemming from the lack of unified optimization for their joint latency, reliability, and resource management.

In-vehicle Ethernet, with its suite of specific protocols, is optimized for in-vehicle data exchange. On the other hand, the Uu air interface uses cellular-based communication protocols for data transfer. Much theoretic and engineering efforts have been paid to bridge the gap between these two distinct protocol stacks. However, the alignment of two networks, in the context of URLLC, presents a complex yet promising landscape. A critical bottleneck lies in reconciling their latency profiles. To be specific, in-vehicle Ethernet is engineered to deliver deterministic latency for in-vehicle applications. By contrast, the Uu air interface's latency inherently fluctuates with network congestion, wireless channel fading, and signal SNR status. Coordinating the URLLC requirements of in-vehicle Ethernet with the varying latency of the Uu air interface is essential for end-to-end URLLC in V2X, motivating the research of network interface controllers, data format harmonization, cross-layer signaling synchronization, resource management, etc.

V. AI FOR V2X COMMUNICATIONS

A. AI AS A 6G V2X TECHNOLOGICAL ENABLER

Driven by advancements in computational capabilities and reductions in cost, AI technology has found widespread application in V2X, particularly in domains like autonomous driving and mobile edge computing. Conventional V2X communication is highly dependent on complex mathematical models. However, these models often lack the necessary precision and accuracy when applied to the highly dynamic and complex conditions of real-world traffic environments. AI technology, particularly deep learning techniques represented by neural networks, has become a core driving force for enhancing V2X communication efficiency, enabling advanced applications, and ensuring system security through its powerful capability to model complex environments. Through these

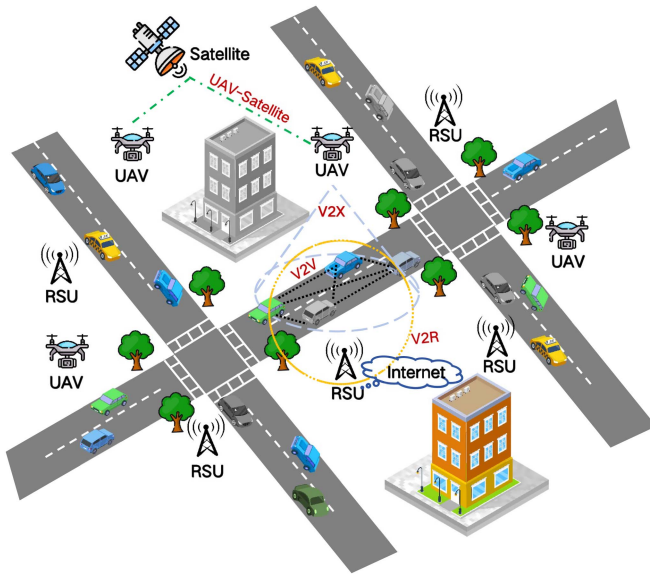


FIGURE 9. The cooperative perception in 6G-V2X ITS system.

technologies, the ITS can directly learn hidden patterns and optimal strategies from massive, high-dimensional real-world or simulated data.

For instance, consider the cooperative perception in ITS illustrated in Fig. 9. In the 6G-enabled space-air-ground integrated V2X network, vehicles can not only acquire local road information through their onboard sensors such as cameras, radar, and Global Positioning System (GPS), but also obtain information and computational assistance from neighboring vehicles, UAVs, RSUs, and satellites via V2X communication. By processing this multi-sourcing and heterogeneous road information, the onboard AI facilitates intelligent vehicle control and decision-making. AI technology has moved autonomous driving into the era of single-vehicle intelligence. Concurrently, due to the surge in computational demands in V2X, research has begun to consider migrating some of the computational and storage resources of traditional V2X to the network edge, closer to vehicles and roadside terminals, in order to provide services to various end-devices through edge computing technology. As a key instrument for resource management, AI is widely applied within this mobile edge computing framework. Furthermore, AI technologies, represented by ML, have also been extensively utilized to ensure the security of data within V2X networks.

Nevertheless, in the 6G era, V2X networks both benefit from the low-latency services brought by high bandwidth and face the challenge of a surge in the variety and dimensionality of nodes and data. Traditional AI-based single-vehicle intelligence cannot support the perception of full spatiotemporal dynamic traffic information and is unable to achieve effective coordination among humans, vehicles, and roads. Furthermore, the need for collaboration between edge, central, and cloud computing poses a significant challenge to intelligent resource management. The generation of vast volumes of

private data also places even greater demands on trustworthy V2X communications.

B. PROPOSALS AND RECENT ADVANCES ON AI FOR V2X

Artificial Intelligence technology uncovers distributions and patterns from massive datasets that are difficult for humans to identify. With the enhancement of computation capability and the reduction of computational costs, the higher throughput and reduced computational latency characteristic of 6G networks provide a foundation for the application of AI in V2X. While 6G networks provide V2X with higher bandwidth and lower latency, a robust communication pipeline alone is insufficient. To fully realize the potential of 6G-V2X, the integration of AI is imperative for enabling efficient perception, intelligent resource management, and trustworthy communication. Table 5 presents the respective applications and advantages of various AI technologies for V2X within the three domains discussed below.

1) AI FOR EFFICIENT COOPERATIVE V2X PERCEPTION

The essence of V2X is to achieve an intelligent road transportation system through coordination among vehicles, pedestrians, and road infrastructure. Efficient cooperative perception is fundamental to coordination, as a reliable basis for subsequent collaboration and optimization can only be established through the precise and rapid sensing of roads, pedestrians, and vehicles. The perceptual data includes not only information from terrestrial sources—such as radar [92], cameras, GPS, and sensors on vehicles and roadside infrastructure—but also data from UAVs and satellites within a space-air-ground integrated network (SAGIN). Given the constraints of limited communication bandwidth, the critical challenge is how vehicles can efficiently share and fuse perception data to achieve perception capabilities beyond an individual vehicle's line-of-sight, enabling the evolution from single vehicle intelligence to a cooperative vehicle-road-pedestrian-cloud intelligence.

One of the mainstream AI approaches for achieving cooperative perception in V2X scenarios is to compress and select features from perceptual data gathered from diverse sources. A representative method in this regard is semantic communication. The concept of semantic communication originates from Shannon and Weaver's definition of different levels of communication [18]. Semantic communication involves the selective feature extraction, compression, and transmission of the original signal. It then utilizes information at the semantic level to communicate, with the ultimate goal of sharing multi-dimensional, multi-modal perceptual data to achieve cooperative perception under limited bandwidth. Unlike traditional source coding that compresses data directly, semantic communication emphasizes the understanding of information, with its encoding and decoding processes depending on a knowledge base trained on large amounts of data and leveraging deep learning networks to handle multi-dimensional, multi-modal information [19]. Numerous studies have focused on the fused perception of multi-source data, including

TABLE 5. Applications of Artificial Intelligence for V2X

Area	Technology	Advantage	Related Work
Cooperative V2X Perception	Semantic Communication	Reducing redundancy and communication overhead and improving communication efficiency	[93], [94], [95], [96]
	Multi-Modal Information Fusion	Overcoming the limitations of individual sensor and improving perception accuracy	[98], [99], [100], [101]
Intelligent V2X Resource Management	Deep Reinforcement Learning	Model-free capability for learning a policy and robust capability for complex environment modeling	[103], [104], [105], [106]
	Graph Neural Networks	Capturing the dynamic topological relationship with strong scalability and graph-learning capacity	[109], [110], [118], [119]
Trustworthy V2X Communication	Traditional Machine Learning	High model interpretability and low computational overhead	[112], [113], [120], [121]
	Advanced Deep Learning	Extracting abstract features and automated data analysis for dynamic new attack detection	[114], [115], [122], [123]
	Federated Learning	Distributed mechanisms and data privacy protection capabilities without sharing raw sensitive data	[116], [117], [124], [125]

images, text, and time-series information, leading to the proposal of various semantic extraction and encoding modules in V2X scenarios [93], [94], [95], [96].

A more prevalent approach for cooperative perception is multi-modal information fusion. By fusing data of different modalities and dimensions perceived by various sensors, the problem of data transmission under limited bandwidth in 6G-V2X scenarios can be addressed at the data source level [97]. Technologies like the Transformer and attention mechanisms can effectively weight perceptual information from various sources, thereby enabling the fusion of multi-source and multi-modal data [98]. Pervasive in autonomous driving, these fusion-based methods integrate data from various vehicle and roadside unit sensors. This strategy concurrently reduces the network bandwidth needed for data-heavy transmissions and compensates for the standalone deficiencies of sensors like millimeter-wave radar, cameras, and Light Detection and Ranging (LiDAR), leading to robust human-vehicle-road cooperative perception [98], [99], [100], [101].

2) AI FOR INTELLIGENT V2X RESOURCE MANAGEMENT

Based on the aforementioned cooperative perception methods, an integrated and cooperative intelligent transportation system involving vehicles, roads, pedestrians, and the cloud can be constructed. As the number of vehicle terminals continuously increases, the massive volume of real-time data generated consequently places enormous processing pressure on the system. Developing dynamic and efficient resource allocation strategies for diverse V2X services remains a critical, unresolved challenge, especially given the need to operate within bandwidth-constrained, computationally-limited, and energy-aware V2X environments that are both large-scale and highly dynamic [102]. The complex and dynamic features of such resource management problems makes them challenging to address with traditional modeling and optimization approaches. Consequently, Artificial Intelligence, particularly deep learning, is widely applied to solve resource management problems in V2X scenarios owing to its powerful modeling capabilities.

Deep reinforcement learning (DRL) is one of the most widely used AI methods in research on achieving intelligent resource management in complex and dynamic 6G-V2X scenarios. deep reinforcement learning (DRL) combines the advantages of both reinforcement learning and deep learning. DRL excels at finding optimal long-term strategies in unknown and dynamic environments through trial-and-error learning. Leveraging its powerful ability to model complex environments, it can handle large-scale, high-dimensional, and uncertain resource management problems in 6G-V2X scenarios, thereby achieving more intelligent resource allocation strategies. DRL is applicable not only to the management of communication resources, enabling dynamic spectrum allocation, but also to the management of computation resources for the intelligent offloading of tasks in V2X [103], [104], [105], [106]. Future work will involve the further application of DRL methods at both the terminal edge and the network cloud. This will enable a hierarchical resource management framework for 6G-V2X networks, designed to support the continuous and robust operation of these complex and dynamic networks.

A GNN is a machine learning model specifically designed to process graph-structured data. It can efficiently learn topological information and features from the graph and possesses strong scalability. Given their aforementioned advantages, GNNs have been extensively applied to address resource management problems within V2X networks, which can also be modeled as graph structures [107], [108]. Through their message-passing mechanism, GNN can fuse local and global node information, thereby better capturing the dependencies between nodes in V2X networks and thus assisting in resource management decisions [109], [110].

3) AI FOR TRUSTWORTHY V2X COMMUNICATIONS

Within 6G-V2X networks, trustworthy communication [111] is a fundamental prerequisite for achieving both efficient cooperative perception and intelligent resource management. Particularly for complex and dynamic 6G-V2X networks, which involve the safety of the entire transportation system

and vast volumes of personal private data, the challenge of how to provide secure and trustworthy communication has consistently been a key focus of numerous studies [17], [111].

Leveraging its powerful feature learning capabilities, machine learning technology is widely applied to the detection of V2X network attacks such as denial-of-service (DoS) attacks, jamming attacks, and replay attacks. For detecting the traditional attacks mentioned above, V2X network communications widely employ detection methods based on various traditional ML models, including support vector machines (SVMs), random forest, K-Means clustering, and Gaussian mixture models (GMMs) [16], [112], [113]. Nevertheless, conventional ML-based security approaches are not readily applicable to the unique environment of 6G-V2X networks. A key limitation is that these methods are often capable of identifying only a specific range of attacks. This specificity makes them difficult to support the demands of complex and dynamic 6G-V2X networks for rapid security responses and trustworthy communication. Furthermore, with the development of AI technology, attacks targeting V2X communications have also advanced considerably. Traditional ML-based methods struggle to identify novel attacks, for example, jamming attacks with AI enhancement and deep-fake attacks on V2X perception [17]. For this reason, a series of V2X network attack detection methods that incorporate advanced AI technologies have been proposed, such as those based on secure reinforcement learning (SRL) [114] and those based on generative adversarial networks (GANs) [115].

Federated learning (FL) is regarded as a key AI technology for addressing the challenge of trustworthy communication within 6G-V2X networks. On the one hand, it allows multiple parties to collaboratively train a model without sharing raw data and can ensure data security and privacy through techniques like encryption and privacy-preserving algorithms. This aligns with the strict requirements of 6G-V2X networks for handling massive amounts of private data. On the other hand, the distributed and collaborative mechanism of FL meets the demand for end-to-end edge intelligence in 6G-V2X networks. The integration of federated learning (FL) and vehicular edge computing (VEC) enables reliable, real-time model training and distributed artificial intelligence, thereby supporting dynamic and distributed collaborative computing services for vehicle-road systems. Many studies have already considered introducing FL into 6G-V2X networks [116], [117].

C. OPEN CHALLENGES AND OPPORTUNITIES ON AI FOR V2X

Although AI technology has provided sufficient strategies for 6G-V2X network communications in the areas of cooperative perception, intelligent resource management, and trustworthy communication, current technologies still face many challenges when confronted with future large-scale, full-coverage intelligent transportation systems. A primary challenge that must be addressed concerns the issues of communication

latency and collaboration efficiency within cooperative perception. Communication delay is an unavoidable problem in cooperative perception. Excessive delay will degrade perception performance and may even lead to results that are worse than those from single-vehicle intelligent perception. Furthermore, even with the support of high-bandwidth, low-latency 6G networks, the inevitable information exchange among agents in V2X creates a trade-off between bandwidth consumption and cooperative efficiency. Realizing end-to-end, real-time decision-making in diverse V2X scenarios is dependent on the converged services of edge, central, and cloud computing. This type of hierarchical computing service places higher demands on resource management. Finally, trustworthy communication serves as the bedrock for V2X applications. This requirement is further amplified in 6G scenarios, where the generation of massive volumes of personal private data necessitates an even higher standard of communication security and reliability.

VI. INTEGRATED SENSING AND COMMUNICATIONS FOR VEHICLES (ISAC-V)

ISAC, as one of the key proposals for the 6G, aims to provide dual communications and sensing functionalities by employing a common transceiver architecture. In the case of vehicles, ISAC becomes even more appealing as the integration of technologies allows to reduce the burden on weight, energy consumption and cost of V2X architectures. More importantly, ISAC allows direct sensing-assisted communications and communications-assisted sensing strategies, contributing to an enhanced development of ADAS for Connected and Autonomous Vehicles (CAVs). Due to its practical relevance for CAVs and to make a distinction w.r.t. generalist ISAC, in this paper we assume the term ISAC-V, introduced in [126], to denote applications of ISAC for vehicular purposes.

The next paragraphs will provide an overview of ISAC-V, review the latest works in the area and identify the main challenges and opportunities to be tackled in the following years.

A. ISAC-V AS A 6G V2X TECHNOLOGICAL ENABLER

Although CAVs heavily rely on both communication and sensing for ensuring safe and efficient operation, traditionally the design of these two technologies has followed separate routes [126]. In order to ensure the safe operation of CAVs, stringent QoS requirements for both sensing and communications are imposed, such as very high throughput (at least 100 Gb/s) and reliability (99.9999 % packet success rate), ultra-low end-to-end delay (lower than 0.1 ms) and centimeter-level localization accuracy [127], [128].

These stringent requirements are not met by current V2X networks [128]. With the advancement of the 5G and the upcoming of the 6G, the paradigm of ISAC emerges in the field of the cellular networks, leading to the so-called Perceptive Mobile Networks (PMNs). ISAC seeks to take advantage of a common set of transmission resources for providing dual

communication and sensing functionalities. Although the research on ISAC for domestic systems started around 15 years ago [129], [130], [131], [132], [133], the concept of integrated communication and sensing can be traced back to the 1960s [129], [134]. Although the most recent denomination is ISAC, references to nomenclatures such as Joint Communication And Sensing (JCAS) [130], Radar-communications (Rad-Com) [135], Joint Communications and Radar (JCR) [131], Joint Radar and Communications (JRC) [134] and Dual-Functional Radar Communications (DFRC) [136] can be found.

By introducing ISAC-V functionalities, PMNs lay the foundations to harmonize the coexistence of mass sensing information from the vehicles' sensors and road-side units, as well as the high-data rates associated to novel on-board services, in already-congested spectrum resources [127]. Hence, joint design of CAVs sensing and communication functionalities has the potential to lead to (a) alleviate the spectrum congestion by using a common signal for both communication and sensing; (b) allow for the direct CAVs sensor data exchange through the V2X network; (c) improving the communication services based on the sensing information (e.g., improved channel estimation, assisted beamforming techniques...); and (d) reducing the size, weight and cost of the CAVs on-board hardware, by integrating both communication and sensing functionalities.

Different from classical monostatic and bistatic definitions [137], [138], the following sensing modes can be defined [126]:

- 1) *Active Sensing (AS)*: The Transmitter Node (TX) and Sensing Node (SX) are the same vehicle or roadside unit, which, while transmitting information to a Receiver Node (RX), senses the surroundings based on the received echoes.
- 2) *Direct Passive Sensing (DPS)*: The RX is also the SX. Hence, the RX, apart from receiving the information from the TX, uses the information from the received echoes for sensing the surrounding environment.
- 3) *Opportunistic Passive Sensing (OPS)*: The SX is a third node not involved in the communication process between the TX and RX. Hence, a silent SX can sense the environment by processing the signal received from the TX or its echoes.

In the 3GPP Technical Report (TR) 22.837 [139], various ISAC-V use cases have been identified for two main applications: automotive maneuvering and navigation and obstacle detection. By considering the 3GPP Technical Specification (TS) 22.137 [140], which corresponds to the DPS mode in [126], 21 different vehicular sensing scenarios are defined in [141]. In order to comply with these vehicular scenarios, a total of 29 communication and sensing architectures are proposed in [141], which can be summarized in the following main classes:

- 1) *Edge-Enabled architecture*: Reduces latency by relying on edge computing entities, namely Edge Enabler Servers (EESs) and Edge Application Servers (EASs), considering solely the transmission of sensing data in the RAN and not the Core Network (CN).

- 2) *Low-latency, Application Function (AF)-friendly architecture*: The sensing is deployed at the RAN without edge computing entities, and the sensing data is provided by an external application, such as a map provider.
- 3) *Low-latency, UE-friendly architecture*: The sensing is deployed at the RAN without edge computing entities, and the sensing data is provided by the vehicle for being analyzed (e.g., for autonomous driving applications).
- 4) *Easy-implementation and High-availability architectures*: The sensing is implemented at the CN, hence not taking any special measure to reduce the end-to-end delay. Since the sensing is implemented at the CN, it may have access to many more resources and support large-scale sensing applications.
- 5) *Wide-coverage architecture*: Architecture designed for when the vehicles are out of the coverage of the RAN.

B. PROPOSALS AND RECENT ADVANCES ON ISAC-V FOR V2X

As a fundamental proposal for future V2X networks, a surge of works on ISAC-V have emerged during the last few years. In the following paragraphs, the most recent advances on ISAC-V are summarized and analyzed from different perspectives. Starting with the topics more related with propagation, ISAC-V channel modeling, sensing parameter estimation, sensing ambiguities resolution, clutter suppression and full-duplex architectures are considered. Continuing with the perspective of the system design, advances on ISAC-V waveforms, security protocols for ISAC-V, resource allocation and network sensing are described. Following the most recent ISAC-V trends, the application of AI and edge computing is considered. Finally, the contributions of sensing on assisting the communications part are detailed. Most of the considered works are also analyzed, by convenience, in Table 6. Note that the aspects on ISAC-V directly related to channel modeling will be analyzed in Section VII.

1) ISAC-V SENSING PARAMETER ESTIMATION

Real-time equalization of the communications channel is generally performed by means of pilot-based strategies, such as in the 4G or 5G, the inclusion of specific preambles, such as in Wireless Fidelity (WiFi), or similar approaches. Although these may be enough to compensate the impairments of the channel for ensuring the real-time transmission of information, they may not suffice in order to infer sensing information about the propagation environment, motivating the need for high-resolution sensing parameter estimation methods [126]. Parameter estimation methods, such as periodogram- or subspace-based methods [142], ray tracing-based approaches for the acquisition of multipath information [143] or multi-dimensional Fast Fourier Transform (FFT)-based methods for the extraction of the velocity, angle and distance of targets [144], have been proposed for ISAC-V applications.

An issue of typical High-Resolution Parameter Estimation (HRPE) methods, such as space-alternating generalized expectation-maximization (SAGE) [182], multiple signal classification (MUSIC) [135] or estimation of signal

TABLE 6. Recent Advances on ISAC-V for V2X

Area	Technology	Advantages	Disadvantages	Related Work
ISAC-V Sensing Parameter Estimation	Pilot/preamble-based methods	Valid for real-time applications	Not applicable for accurate sensing parameter estimation	–
	Subspace-based methods	- High resolution - Ability to extrapolate estimates	- High computational complexity - Dependent on adequate signal models - Require consecutive samples	[142]
	Ray tracing-based methods	Accurate results	- High complexity - Require certain degree of knowledge of the propagation environment	[143]
	Multi-dimensional FFT-based methods	Considers the stochastic components of the propagation process, enabling higher adaptability	Challenges not yet addressed: in-band interference and features set selection for detection and tracking of targets	[144]
	Learning-based methods	- Once trained, computationally efficient - Suitable for complex scenarios	- High computational cost for training - Lack of adaptability without re-training	[145]
	Compressed sensing	- Flexible and adaptable - Not requiring large amounts of continuous samples	- High computational complexity for some methods - Require good separation between different parameter values	[146]
Resolution of Sensing Ambiguities	Cross-Antenna Cross-Correlation	- Single-link required - Applicable to standard WiFi links	Strong assumptions: prominent LoS component and static UEs and BSs	[147], [148]
	High-Resolution Parametric Estimation	- High-resolution results - Reduced set of parameters	Computational cost	[147]
	Inter-antenna CSI ratio	- Immunity to measurement noise - Higher information provided w.r.t. CACC	Challenge on separating signals from different sources	[149], [150]
Clutter Suppression	Classical radar techniques	Mature algorithms	Only specific scenarios (clutter with certain features, few targets in open space...)	[151]–[154]
	Background suppression	Well-known image-processing methods can be extended for ISAC-V	- Sensitive to time/frequency offsets - Generally limited to separate static/dynamic targets	[132], [155], [156]
Full Duplex Integration	Communications-centric Full-duplex	Nearly doubling communications data rate	Hardly applicable in dynamic environments	[157]–[159]
	Full-duplex ISAC by static/dynamic targets separation	Good performance for moving targets	Still challenging for static targets due to self-interference	[160]
	Full-duplex ISAC by multi-beam strategies	Reduction of leakage and clutter by separating communication and sensing beams	Still challenging to be applied to MIMO ISAC-V	[130], [161], [162]
ISAC-V Waveform Design	OTFS	- Representation in the DD domain, with direct application to ISAC - Circumvents Doppler spread effects, being a natural ISAC-V waveform candidate	Self-interference in pilots by data symbols	[163], [164]
	OTFS with guard pilots	Self-interference reduction	High-pilot overhead	[165], [166]
	ST-OTFS	Mitigation of pilot overhead	Increased computational complexity	[167]–[169]
Secure ISAC-V	Analysis of open problems	–	Several challenges yet to be tackled: sensing authentication and authorization, sensing data integrity and encryption, and sensing consent management by targets	[141], [170]
ISAC-V Resource Allocation	Joint optimization of sensing/communication metrics	Allow to ensure QoS simultaneously for sensing and communications	Practicality of current proposals on resource allocation schemes not proven	[127], [171]
Networked Sensing	–	Distributed nature of PMNs inherently allows for distributed and cooperative sensing	- ISAC algorithms exploiting PMN architectures not available yet - PMNs challenges not addressed yet, e.g. cell interference or handovers for sensing	–
AI/Edge Computer-empowered ISAC-V	UAV sensing aided by Edge Computing	Minimizes UE consumption while maximizing accuracy by training an AI model at the edge-server	Multiple-UAV sensing not yet supported	[172]–[176]
	AI-aided distributed ISAC	Enables high-quality cognitive V2X services	Heavy computational burden, requiring edge-computer empowered strategies	[170], [172]
	AI-aided ISAC-based ADAS	- Decentralized sensing strategy - Empirically proven practicality and accuracy	- Affected by terrain blockage - Not able to deal with multiple vehicles in different trajectories	[177], [178]
Sensing-assisted communications	Sensing-assisted beamforming	- Natural application of co-location and co-frequency of sensing and communication solutions - Especially relevant for dynamic transceivers	Most sensing approaches consider targets as a punctual object located at the corresponding reference antenna element	[127], [179], [180]
	Eavesdroppers avoidance	Allows for SNR minimization at eavesdroppers while ensuring a SINR threshold at the target	Secrecy reduced in realistic scenarios (target location uncertainty, non-perfect CSI knowledge...)	[181]
	Secure PHY secret key generation	Increased secrecy w.r.t. CSI-based methods	No actual solutions proposed yet	[129]

parameter via rotational invariance technique (ESPRIT) [183], for their applicability to sensing parameter estimation is that they work properly when the samples are uniformly distributed in different domains (e.g., time or frequency). However, considering the complex structure of the typical frames for current communication systems, such as 5G or WiFi, it is difficult that will be the case. On the other hand, the required sensing rate for practical applications is typically in the order of milliseconds to seconds [126], [129], which may be affordable from a computational point of view in a BS (e.g., for an Edge-Enabled Architecture, as specified in Section VI-A), but it can hardly be applicable to situations in which both ends are mobile, such as V2V scenarios. For these cases, methods based on learning, in which the computational burden is performed offline [145], or compressed sensing-based approaches [146], can be considered.

An initial approximation to a complete framework for sensing channel parameter estimation focused on the characteristics of the targets can be found in [144]. Table 6 summarizes the main advantages and issues of the application of different channel parameter estimation techniques for ISAC-V.

2) SENSING AMBIGUITIES IN CURRENT PMNS

Especially for the DPS and OPS sensing modes, a problem related to the architecture of current cellular networks arises: timing offset and Carrier Frequency Offset (CFO) between the UEs and BS, which will lead to ranging and speed ambiguities, respectively [126], [129]. This problem is even more remarkable for highly-mobile UEs, as expected in V2X networks. Under some specific circumstances, compensation methods have been reported in [147], [148], [149], [150] by using the Cross-Antenna Cross-Correlation (CACC), HRPE methods or exploiting the ratio of CSI measurements across antennas.

3) CLUTTER SUPPRESSION

Typical V2X environments involve complicated propagation scenarios leading to rich-multipath environments. The Multi-Path Componentss (MPCs) received at the sensing side are in general useful for communications, as they contribute to diversity, but not all of them are of interest for sensing. For example, in V2I AS architectures, being a static BS the sensing node, MPCs originating in moving targets can be of interest, whereas permanent long-term MPCs originated in large architectural elements bear little information and are usually referred to as *clutter*.

Although clutter suppression techniques are not a new concept for classical radar, typical approaches such as [151], [152] may not be directly applicable to ISAC-V, as they are optimized for narrow beams and in environments where clutter has distinctive characteristics from targets (e.g., reflections on the ground or the sea surface) [151], [153], [154]. Background subtraction methods have also been proposed for clutter removal in PMNs [129], [132], [155], [156].

Due to clutter suppression in typical V2I environments can be equaled to distinguishing dynamic targets from long-term static targets, methods for separation of moving and static targets, such as [184], [185], can serve for that purpose. However, it must be noticed that these approaches may not be valid in all circumstances. For example, in V2V applications, two vehicles moving in parallel lanes could detect each other as clutter.

4) FULL-DUPLEX NECESSITY

When considering the AS sensing mode, another classical problem is the signal leakage. Although the problem can be partially alleviated if only co-located sensing at the TX is needed (and without simultaneous transmission between communications nodes [129]), a general solution would need the implementation of full-duplex solutions. Although full-duplex structures have widely been considered for communications [157], [158], [159] and some incipient approaches have been considered for full-duplex ISAC [130], [161], [162], it still remains a challenge to implement full-duplex ISAC structures in mobility, especially for MIMO systems.

5) ISAC-V WAVEFORM DESIGN

Traditional radar and communication systems tend to use very different waveforms (e.g., unmodulated for the first and random multi-carrier modulated for the later), whose optimization criteria are often opposed. If a common signal is to be used for both tasks compromises are to be made.

OFDM is undoubtedly the leading waveform in most modern mass-scale communication systems (4G, 5G, WiFi...) due to its high spectral efficiency, easy equalization and immunity to selective fading. These appealing advantages compensate its shortcomings, such as its high Peak-to-Average Power Ratio (PAPR) or sensitivity to the Doppler frequency in high-mobility scenarios.

As a matter of fact, it is well-known in the field of vehicular communications that Doppler shifts, reducing the channel coherence time, destroy the orthogonality of OFDM, leading to Inter-Carrier Interference (ICI). With the upcoming of ISAC, the questioning of the well-established OFDM and, in general, the design of waveforms more adequate for the new application scenarios, such as sensing in vehicular applications, came again into scene. Orthogonal Time Frequency Space (OTFS), first introduced in 2017 [186] based on results that date back to the 1960s [187], emerged first as an alternative for high-mobility applications, to recently be almost unanimously recognized as an excellent candidate for ISAC and ISAC-V [186], [188], [189].

The OTFS basis is to modulate data in the Delay-Doppler frequency (DD) domain, rather than the traditional Time-Frequency (TF). On the one hand, this circumvents the time-varying effect caused by the Doppler spread. Moreover, the unitary transformation from the DD domain to the TF domain spreads each symbol along the entire TF domain, leading to full-frequency diversity [163], [164]. In the case

of ISAC, given that propagation delay is directly related to range estimates and Doppler frequency to target speed estimates, sensing can be directly performed in the OTFS DD representation. Hence, OTFS poses as a good candidate for ISAC and, given its resilience in high-speed scenarios, for ISAC-V in particular.

Despite the great advantages of OTFS for ISAC-V, some challenges are still to be solved, as it has been reported that OTFS demands a large pilot overhead, since each pilot must have a set of guard symbols surrounding it to avoid self-interference caused by the data [165], [166]. In order to combat this problem, Superimposed Training (ST) has been proposed [167], [168], [169], leading to the so-called Superimposed Training-OTFS (ST-OTFS), based on a superimposition of pilots and data, which inherently causes self-interference. A ST-OTFS framework for performing joint data transmission and sensing capable of reducing the self-interference is proposed in [167].

6) SECURE ISAC-V

ISAC-V solves a typical concern of camera-based vehicular systems, namely the privacy issues [145], [190]. However, ISAC-V is not exempt of security challenges. These challenges incur in the different phases of the sensing process: the initiation or request of the sensing, the transmission and storage of information and privacy and user consents regarding the obtained sensing results. In particular, the following key points are identified [141]:

- *Sensing authentication and authorization:* In order to avoid privacy issues for targets, sensing services have to be authenticated and sensing authorizations need to be implemented. Unfortunately, this goes against the stringent timing constraints inherent to many ISAC-V services. In such cases, sensing tokens or other pre-authorization approaches can be considered.
- *Sensing data transmission and storage:* Integrity of sensing data transmission is of utmost importance. Malicious vehicular sensing data alteration can lead to accidents. Most data integrity protection methods operate in layers higher than the PHY [141], not offering protection for sensing data transmitted at the PHY. Furthermore, although encryption of data at the network level is possible before the transmission, the identity and location of the sender could still be leaked by just detecting the existence of a communication link [170].
- *Data privacy and user consent:* Consent procedures for sensing (by the target) should be implemented prior to the sensing process occurs. Authorization procedures such as these in 3GPP TR 33.867 [191] can be implemented e.g. in V2V or V2I applications, where both ends can execute the required protocols. However, when targets are humans, e.g. a vehicle sensing people, these kind of protocols are not applicable. Furthermore, timely deletion of sensing data is essential to ensure privacy.

7) ISAC RESOURCE ALLOCATION

Proper resource allocation is a key requirement to guarantee acceptable QoS levels for different concurrent services and users. However, when the services considered involve different figures of merit (e.g., sensing and communication) optimization of resource allocation techniques becomes a challenge. In [127], [171], an approach considering an AS architecture in which different pencil-sharp narrowbeams are used for different services (communication, localization, detection and ISAC-V) and users. The key is to allocate the resources by optimizing the most critical metrics for each service, including sensing metrics such as probability of detection, Cramér-Rao Bound (CRB), Posterior CRB (PCRB) and communication metrics such as the achievable sum-rate [127]. In [171], a convex problem is formulated based on these criteria in order to optimize the ISAC resource allocation.

8) NETWORKED SENSING

The inherent distributed nature of cellular networks brings us with the opportunities of distributed and cooperative sensing. However, there are not known algorithms yet exploiting the specific characteristics of cellular networks and some challenges have to be addressed yet, such as interference management between cells or cell-handover for sensing applications. Besides, most works in the literature consider the AS mode, which may not be adequately adapted to a multi-node scenario.

9) AI AND EDGE COMPUTING-EMPOWERED ISAC-V

Several of the sensing architectures presented in Section VI-A consider the latency reduction by using edge computing resources. Recently, an incipient surge on works of edge computing-aided ISAC-V, especially for UAV sensing applications, can be appreciated, with several recent works published, such as [172], [173], [174], [175], [176]. In these works, not only limited to ISAC, mobile edge computing serves for optimizing scheduling, energy consumption and offloading strategies, or for offloading data for training models on tracking or identification of targets. Given the generalist definition of edge computing architectures in 3GPP TS 23.558 [192], it is expected that more edge computing-empowered ISAC-V applications will surge in the near future, especially for very low-latency services or for computation/energy-constrained vehicles (e.g., UAVs).

On the other hand, as considered in some of the works (e.g., [172]) the massive amount of data generated by ISAC-V nodes at distributed locations opens the door to high-quality cognitive services for V2X networks [170] when aided by AI models. Studies have already shown the viability of vehicular head-on-collision warning systems by applying machine learning models to decentralized radio sensing signals [177], [178].

10) SENSING-ASSISTED COMMUNICATIONS

A side-advantage of ISAC systems comes specifically due to the integration of both sensing and communication functions. In particular, the usage of sensing information to benefit the communications has been proposed. For example, a radar in a BS, collocated with the communications equipment, is proposed in [179] for communications beamforming. However, since the considered bands are disjunct, the results are not optimum. ISAC by inherently using a common transceiver and signal for both communications and sensing, offers a more adequate approach. In the particular case of vehicular applications, due to the mobile nature of the transceivers, sensing-assisted beamforming becomes an even more interesting application of ISAC-V, with incipient works, such as [127], [180]. One aspect yet to consider, especially when not constraining the sensing to punctual scatterers (see Section VII-B2) is that general sensing is aimed to detect or track objects (e.g., vehicles) more than their specific transceivers or antennas onboard [129].

Although itself ISAC-V poses severe security issues (see Section VI-B6), the counterpart is that ISAC-V-assisted adaptive beamforming techniques can be used to increase the security of communications. For example, it is possible to combat eavesdroppers by inserting artificial noise whereas formulating an optimization problem to minimize the SNR received by eavesdroppers and guarantee the required Signal-to-Interference-plus-Noise Ratio (SINR) at legitimate users [181]. Sensing information can also be used to increase the secrecy capacity of PHY key generation methods w.r.t. methods based on CSI [129].

C. OPEN CHALLENGES AND OPPORTUNITIES ON ISAC-V FOR V2X

Despite the arguable advantages of the convergence of sensing and communication functionalities for vehicular networks and the recent works on the matter, summarized in Section VI-B, there are still challenges to be solved before ISAC-V becomes a reality. The identified challenges have been divided into propagation-related, system design-related and recent trends-related challenges.

1) PROPAGATION-RELATED ISAC-V CHALLENGES

Inaccuracies on current ISAC-V channel models impose great limitations to practical applications, as it will be analyzed in Section VII-B2. However, even the ISAC-V channel models were adequate, estimating their parameters to perform the actual sensing incurs in several challenges as well. On the one hand, although a pilot/preamble-based channel estimation is enough in general for performing channel equalization at the receiver side, it cannot provide estimates with sufficient resolution for sensing scatterers. Hence, more complicated approaches, such as HRPE estimators are needed, but these come at a high computational cost, which goes in detriment

of the stringent time-requirements for V2X sensing applications. On the other hand, typical HRPE methods require some sort of uniformity of the samples used for sensing in some domains (e.g., time and frequency). However, this uniformity can seldom be met for current signal structures, such as 4G, 5G or WiFi. Finally, in order to separate targets from clutter, most of the current methods perform a simple dynamic/static scatterers classification. However, the targets of interest are not always dynamic in V2X scenarios.

It is worth noting that for general AS deployments, full-duplex is a requirement. However, although full-duplex architectures have been considered, it still remains a challenge to implement full-duplex ISAC in mobility, especially for MIMO systems.

2) SYSTEM DESIGN-RELATED ISAC-V CHALLENGES

The weaknesses of OFDM in vehicular applications are well known. Although OTFS poses as an excellent ISAC-V 6G waveform candidate due to the direct sensing interpretability of the DD domain and its resilience against vehicles' speed, it introduces a notable pilot overhead, which can only be reduced by increasing the self-interference.

On the other hand, critical security problems need to be addressed in regards to ISAC-V applications. The nature and time-requirements of sensing applications makes difficult or impossible to consider typical methods to ensure the security and privacy of sensing: not all receivers can implement consent protocols and the sensing time-requirements may not allow for sensing authorization processes. Furthermore, a malicious alteration of the sensing information can lead to accidents.

How to balance communication and sensing resources, as well as how to combine distributed sensing resources, is also a challenge. On the one hand, the sensing quality metrics are not as standardized as the ones for communications. Allocating resources for both services to fulfill the QoS requirements when different quality metrics are used is not trivial. On the other hand, despite the appealing advantages of the cellular structure for cooperative sensing, algorithms exploiting cooperative PMN strategies between different nodes are not yet available.

3) RECENT TRENDS-RELATED ISAC-V CHALLENGES

The massive data generated by multiple sensing nodes fits perfectly for AI-enabled ISAC-V strategies. However, these incur in large computational costs and they may exhibit limited generalizability. Although edge computing-empowered ISAC-V could be an alternative to alleviate the computational cost, no actual architectures are standardized yet for this purpose.

VII. V2X CHANNEL MODELING

Channel models are widely-recognized as one of the key pillar for the optimization of transceivers and network deployments in different applications [193]. V2X communications are not

the exception, and hence channel models and channel modeling techniques are essential to provide the required physical support to most of the identified key V2X technological enablers, such as NTN or ISAC-V.

The next paragraphs will provide an overview of the key aspects of V2X channel modeling, review the latest works in the area and identify the main challenges and opportunities to be tackled in the following years.

A. CHANNEL MODELING AS A 6G V2X TECHNOLOGICAL ENabler

V2X communications are quite a challenging component of mobile and wireless communications, since they encompass a very diverse range of deployment scenarios, usage cases, and supported services and applications, hence, leading to a very diverse set of requirements. This wide set of requirements has implications on how the actual systems and networks should be designed, e.g., with a stricter or a looser view on system reliability, allowing for a network to work with a lower or a higher probability for failure; for example, for some demanding cases, the maximum failure allowed for the network is 99.999 % (a bit more than 5 minutes per year), which is the one taken as a reference for 5G [194], but for much less demanding ones it can go down to 99.9 % (almost 9 hours per year). Given that this reliability is seen on the whole usage perspective, i.e., the user not being able to use the network for the quality of service that has been contracted, the implications are that the full chain in the system and the network components needs to take this into account, ranging from the communication between the mobile terminal and the base station, to the queuing of packets in the application server. The channel model plays a key role in the design of the communication between the mobile terminal and the base station, which is why one needs models that describe as accurately as possible all propagation phenomena in the targeted usage scenario, so that reliability goal can be achieved. Moreover, when ISAC-V enters into scene, the channel is no longer exclusively related to the design and optimization of communication systems, but for radar and sensing as well.

It is well known that the radio channel model, which accounts for the characteristics of both transmitter (Tx) and receiver (Rx) antennas (antennas models) and wave propagation (propagation channel model) [195], is an increasing component of systems' air interface characterisation. When systems had just passive antennas on both ends (Tx and Rx), which happened basically until 4G, the study of the propagation channel was enough to provide a good model for the characterisation of the propagation scenario, but these days, when active antennas exist in at least one of the ends of the radio link, which is the case for 5G in the 3.6 GHz band, one does need to develop the full radio channel model. On the antennas side, the characterisation of the radiation patterns of the antennas, namely addressing beamforming and beamsteering (the latter when existing), as well as Massive MIMO (Multiple Input Multiple Output) aspects, are essential for the development of such model [196] (but others should not be

forgotten, such as polarisation [197]); on the propagation side, well known aspects of wave propagation need to be accounted for, decomposing a quite complex phenomenon into path loss and fast and slow fading components [198].

The radio channel model should also account for other aspects, e.g.: the vehicle's speed (which impacts on a number of features, e.g., coherence time); the pace of change of environment (e.g., from a street to an underground parking, which impacts on a number of features, e.g., coherence band); and the materials of the surrounding environment and their distance to the vehicle (e.g., mostly metal close to the vehicle in a parking place or mostly dielectric away from the vehicle in a large urban square). In addition, the type of antennas on both ends of the link need also to be accounted for, e.g., see [197]. Although these days, one can say that active antennas (as well as Massive MIMO) exist only on the base station, one can state that vehicles offer much more degrees of freedom to deploy antennas than phones or tablets, hence, opening the possibility to actually deploy active antennas on them with addressing beamforming and beamsteering capabilities. However, design constraints of vehicle manufacturers will have a say on antennas' location and performance, which also need to be accounted for (many years ago, broadcast radio antennas cars would be a whip on the car rooftop, but currently they are mostly in the rear window glass).

Furthermore, V2X can be decomposed into several cases, namely [199]: V2V, V2I, V2N and V2P. These various cases put additional challenges in the characterisation of scenarios, since certain characteristics can be quite different, e.g.: radio link distances; the type of antenna on the other end of the radio link; number of simultaneous radio links; speed of change of the link characteristics. These cases also mean that the variety of services and applications they will support will be quite different. 3GPP has established a list of service requirements to be fulfilled by V2X communications in 5G, for 6 areas: General Aspects - interworking, communication-related requirements valid for all V2X scenarios; Vehicles Platooning; Advanced Driving; Extended Sensors; Remote Driving; Vehicle quality of service Support. The requirements include several parameters, such as: maximum end-to-end latency; reliability; data rate; and maximum communication range. A careful analysis of these parameters allows to identify the importance of radio channel models: distance plays a direct key role in latency and range, and an indirect one in reliability and data rate; the former is exempted from an explanation, but the latter can be understood when one considers that data rate monotonically depends on SNR, at least for Quadrature Amplitude Modulations (QAM), hence, on distance, and that reliability is a function of SNR, thus, again, on distance. Radio channel models allow the characterisation of the received signal as a function of distance, among other parameters, such as, frequency and time in general terms, and geometrical and other characteristics in terms of the environments and of both ends of the link [200]. It is apparent then that there is a need for channel models for all environments, scenarios, usage cases, and services and applications to be considered for V2X.

B. PROPOSALS AND RECENT ADVANCES ON CHANNEL MODELING FOR V2X

As a fundamental basis for future V2X networks, a surge of works on V2X channel modeling have emerged during the last few years. In the following paragraphs, the most recent advances on V2X channel models in the key areas directly affected by propagation from these identified in this paper (NTN and ISAC-V) are summarized and analyzed from different perspectives. For convenience, Table 7 includes an overview of the channel modeling works included in this section.

1) NTN CHANNEL MODELING

As introduced in Section III-A, two main platforms are considered under the umbrella of NTN, namely spaceborne and airborne systems [49], the latter mostly dominated by the UAVs. Due to the increasing interest of the integration of aerial nodes into the 6G, a surge on related works on channel modeling arose.

For the case of the HAPS, and more specifically UAVs, the interest started already with the 5G, when the 3GPP approved the corresponding study and work items [247], [248] to investigate the feasibility of serving UAVs by terrestrial LTE deployments [218]. The main differential characteristics of UAV Air-to-Ground (A2G) channels w.r.t. purely terrestrial channels can be summarized as follows [218]:

- Terrestrial BSs are focused on serving ground users, and hence downtilted. The ground and the architectural elements naturally limit the area of each cell, and therefore constrain the interference to neighboring cells. This downtilted configuration is far from optimum to serve aerial users like UAVs. At the same time, if radiation lobes pointing upwards were used, intercell interference would be introduced due to the lack of clear blockages in the air [249]. Moreover, the UAVs themselves would cause interference to other UAVs and to terrestrial users as well [202].
- Although one could be tempted to assume close-to-Line-of-Sight (LoS) propagation for A2G UAV channels and consider traditional aeronautical channel models, these are mostly designed for large airplanes at much larger heights and with dedicated and elevated ground towers. However, since UAV flights are performed in general at much lower altitudes, the terrestrial objects and architectural elements introduce non-negligible effects on the propagation channel. These lead to clearly distinctive characteristics from these observed in traditional aeronautical channels, such as a non-monotonically increasing K-factor with the flight height, as reported in [218].

Different approaches for the characterization of the A2G communications channel for UAVs are proposed in the literature. In the next paragraphs the main related works are analyzed.

Simulation-based works have been considered by some authors. For example, a combination of the 3GPP channel models in [201] for low-height flights and free-space

propagation for higher altitudes has been considered in [202]. Ray-tracing has been also USED to study the influence of the elevation angle on the path loss and shadowing [203], as well as to characterize UAV A2G mmWave propagation in different environments [204].

Being the path loss one of the most relevant channel characteristics, several works have provided measurement-based characterizations of it for different scenarios, flight heights and carrier frequencies. For example, LTE deployments are considered in [205] in the band of 850 MHz in typical suburban environments for flight heights between 15 m and 120 m. Similar flight heights have been considered in [206] for open field and campus scenarios. The authors have also included the effect of the UAV orientation in their proposed models.

Shadowing and small-scale fading have also deserved attention in empirical models. In [207], the shadowing, together with the path loss, is characterized for A2G LTE communications for flight heights between 1.5 m and 120 m, whereas [208] characterizes both the large- and small-scale fading for very low flight heights (4 m to 16 m) in Ultra-WideBand (UWB).

In [209], the variation of the mean Angle of Arrival (AoA) and Angular Spread (AnS) with the flight height is evaluated based on experimental measurements in both urban and rural scenarios using commercial LTE deployments. The work was extended in [210], where comprehensive spatial channel characteristics were investigated in the composite and cluster levels at different heights.

By considering a piloted aircraft at the air side, being the flight height values between 500 m and 2000 m and the link distances between 500 m to 50 km, D. Matolak, et al. thoroughly characterized the A2G channel characteristics. Path loss and MPCs were characterized in different scenarios, including over-water [211], [212], over-freshwater [213], hilly and mountainous environments [250], and suburban scenarios [214], [215], including hilly suburban environments [216]. In [222], a complete channel model including Power Delay Profile (PDP), Doppler frequency Delay Profile, mean delay, Root Mean Square (RMS) Doppler frequency spread and RMS delay spread for different flight phases (in-route cruise, climb-and-descent and takeoff-and-landing) was proposed.

Several contributions on A2G and Air-to-Sea (A2S) statistical channel modeling were introduced by Rodríguez-Piñeiro et al. In [220], [221] a stochastic channel model was proposed, including characterizations of path loss, shadow fading, delay spread and Doppler frequency spread for horizontal and vertical flights at different heights and distances to the BS. A more detailed stochastic model for the same scenario including different types of antennas and the effect of partial blockage from architectural elements was proposed in [218], concluding that many of the assumptions taken in previous A2G UAV works are invalid. In [217], the previous model was extended to provide a spatial-consistent channel modeling approach resistant to blockage by architectural elements, whereas [219] established a time-varying channel model based on the

TABLE 7. Proposals on Channel Modeling for V2X

Area	Category	Characteristics	Related Work
NTN (Airborne, UAV A2G)	Hybrid 3GPP-based simulators	Combination of 3GPP channel models in [201] for low heights with free-space propagation for high-altitudes	[202]
	Ray tracing-based models	- Scenarios: urban, suburban, rural and over sea - Frequency bands: mmWave - Additional characteristics: consideration of the dependency with elevation angle	[203], [204]
	Measurement-based path loss characterization	- Scenarios: suburban and open field - Frequency bands: 850 MHz - Flight heights: 15 m–120 m	[205], [206]
	Shadowing and small-scale fading	- Scenarios: open area and blockage by foliage - Frequency bands: 800 MHz / UWB (3.1 GHz–5.3 GHz) - Flight heights: 1.5 m–120 m / - Link distances: 1 km–22 km	[207], [208]
	Angular parameters	- Parameters: AoA, AnS, spatial channel characteristics at composite and cluster levels - Scenarios: urban, rural and industrial - Frequency bands: 1.8 GHz - Flight heights: 0 m–40 m	[209], [210]
	Stochastic channel characterization (high flights)	- Parameters: path loss, delay spread, stationary distance, K-factor and interband/spatial correlations - Scenarios: over-water, hilly, mountainous, suburban, near-urban and hilly suburban. - Frequency bands: L- and C-bands (bandwidths between 5 MHz and 50 MHz) - Flight heights: 500 m–2000 m - Link distances: 500 m–50 km	[211]–[216]
	Stochastic channel characterization (low flights)	- Channel models: standard stochastic channel model, spatial-consistent channel model and time-varying channel model - Parameters (stochastic models): Path loss, shadowing, PDP, Doppler frequency PSD, delay spread, Doppler frequency spread, K-factor - Parameters (spatial-consistent models): instantaneous MPCs, short-term MPCs and long-term MPCs - Parameters (time-varying models): MPC survival length, distance between consecutive MPC births, initial delay and distance of MPC births, linearized model for MPC trajectories, number of simultaneous alive MPCs and correlations between the previous parameters - Scenarios: Suburban (LoS and OLoS). - Frequency bands: ≈ 2.5 GHz (bandwidths around 10 MHz) - Flight heights: 15 m–105 m - Link distances: 20 m–500 m	[217]–[221]
NTN (Airborne, UAV-USV A2S)	Stochastic channel characterization	- Parameters: PDP, Doppler frequency delay profile, mean delay, RMS Doppler frequency spread, RMS delay spread, ranging accuracy - Scenarios: in-route cruise, clumb-and-descent, and takeoff-and-landing. - Frequency bands: L-band (10 MHz bandwidth) - Flight heights: 3 km–9 km (30 m–330 m for takeoff and landing cases) - Link distances: 500 m–350 km	[222]
	Stochastic channel characterization	- Parameters: path loss, shadowing, multipath fading - Frequencies: up to 2.2 GHz (bandwidths around 6 MHz) - Flight heights: 50 m - Link distance: up to 3.1 km	[223]
NTN (Spaceborne, Satellite A2G)	LMS fading models	- Scenarios: open area, shadowed, foliage blockage (heavy and light) - Frequency bands: L-band	[224]–[226]
	LMS wideband fading models	- Parameters: amplitudes, phases, PDPs, Doppler frequency PSDs, inter-tap cross-correlations - Frequency bands: L-, S- and Ka-bands - Additional characteristics: consideration of the dependency with elevation angle	[227]–[230]
	Clutter loss modeling	- Types of models: empirical, hybrid ray tracing & empirical, semideterministic based on ray-tube effect - Frequency bands: 1.5 GHz–14 GHz (mostly)	[231]–[239]
ISAC-V	Geometry-Based Stochastic Models	- Conventional channel model for communications part - Locations of multi-bounce scatterers are determined - Only fixed-to-mobile scenarios - Large antenna arrays required	[240]
	Geometry-based Spatial Consistent MPC-based Models	- Able to detect scatterers in the environment - Able to discriminate “ghost” targets due to multi-bouncing - Only punctual scatterers considered	[217], [241]
	Trajectory-based Physical Target Discrimination	- Fully compatible with multi-bouncing scenarios - Compatible with targets with geometrical extension (not punctual scatterers) - Manual thresholds needed - Challenging for non-uniformly moving targets	[242]
	Multi-link approaches	- Separated clusters for communications and sensing, some of them shared - Stochastic characterization only applicable to the communications part	[243], [244]
	Hybrid approaches	Combination of AI and ray tracing to predict mobility profiles	[245]
	Geometry/kinematic characteristics-based methods	- Channel characteristics related to the geometric/kinematic characteristics of the targets and their dynamic observability - Provide stochastic characterization of the sensing part of the channel	[246]

stochastic characterization of the variation of the MPCs. Finally, [223] consider the stochastic characterization of the propagation between a UAV and a Unmanned Surface Vehicle (USV).

Different from the case of the HAPS (especially low-height UAVs), when space-borne platforms (generally satellites) are considered, the channel characteristics are dominated by the huge propagation delay and power loss of the signals. The specific geometry of satellite A2G propagation leads to noticeable differences in other channel characteristics as well, e.g. the angular spread is close to zero [227], whereas this assumption is unrealistic in terrestrial deployments given the limited height of the BSs [239]. As a matter of fact, the Land Mobile Satellite (LMS) channels have received attention for decades. In [224], the LMS channel at L-band has been characterized for both directive and omnidirectional antennas. A fading model is proposed with two possible states, namely open and shadowed area. [230] presents wide-band characteristics for low-elevation LMS MIMO channels at S-band in three environments (tree-lined road, suburban and urban). First and second order statistics and inter-tap cross-correlations are derived for both large and small scale fading. In [225], a LMS shadowed Rice model with Nakagami LoS component is proposed. The model leads to closed-form and mathematically-tractable expressions for the fundamental channel statistics, enabling easy optimization of transceivers or network deployments. In [226] the LMS channel at 2 GHz is measured and the signal envelope is stochastically characterized for two different scenarios, namely the heavy and intermediate tree shadowing cases. In [229], a statistical model for the LMS channel capable of describing narrow- and wide-band conditions is presented. The model, applicable to both GEO and NGSO satellites, can produce time-series of amplitudes, phases, instantaneous PDPs and Doppler Power Spectral Densities (PSDs). Empirical parametrization of the model for different environments and elevation angles is provided for L-, S- and Ka-bands. Standardized models have also been proposed by the 3GPP [227] and the ITU Radiocommunication Sector (ITU-R) [228].

As it was already discussed for the case of the UAVs, the influence in the channel characteristics of the architectural elements, vegetation and other terrestrial objects must be considered if we want to ensure seamless and ubiquitous A2G communications [218]. For the case of satellite A2G communications, this aspect becomes even more critical due to the possibility of facing very low elevation angles. Despite its practical relevance, the attenuation from the terrestrial elements, namely the Clutter Loss (CL), is not comprehensively included, but in turn relegated to a few empirically-obtained values, even in the most recent models [227], [228]. The ITU-R P.2108 recommendation [231] aims to address this gap, but it still includes many empirical parameters binded to certain experimental scenarios, as well as it exhibits lack of interpretability. A few works have recently addressed the CL characterization following different strategies. In [232], the CL for A2G channels at 5.8 GHz is evaluated and a

modification of the ITU-R P.2108 recommendation [231] for low-frequency bands is proposed. Another model based on the same ITU-R recommendation is proposed in [233] for the CL at the 1.5 GHz, 3.3 GHz and 5.7 GHz bands for both LoS and building-obstructed scenarios. The model is extended to higher frequencies in [234] and for different ground node heights in [235]. The combination of ray-tracing simulations and field measurements led to the derivation of a CL model considering the elevation angle and the height of the ground node for frequencies below 10 GHz in [236]. The effect of the elevation angle and the CL was also preliminarily studied in [237]. Several of the works lack of interpretability and ignore the promising frequencies in the range of 6–24 GHz, which are recognized as promising for the 5G-Advanced and 6G systems [238]. In order to fill in these gaps, in [239], a concise semideterministic model for the CL in NTN is provided. The model constitutes an interpretable framework relating the CL with the aerial platform elevation, by introducing the variation of the power flux density from the ray-tube effect inherent to NTN channels.

2) ISAC-V CHANNEL MODELING

Although significant efforts on ISAC channel modeling have been appreciated recently, most of them are inherently designed for static scenarios. Among the works on channel modeling for ISAC-V, most of them correspond to the AS sensing mode, with no relevant works identified for DPS and OPS [126].

In [240], a non-stationary fixed-to-mobile (F2M) ISAC geometry-based stochastic model is presented, where a conventional channel approach is considered for the communications part. Through the combination of communication and sensing, the locations of multi-bounce scatterers in the channel are determined. The authors assume the provision of large antenna arrays at both ends of the communications link in order to extract the angular parameters of the channel.

In [217], the Geometry-Based Spatial-Consistent MPC Tracking Method (GSTM) framework, able of tracking the evolution of the MPCs of vehicular channels is introduced. By exploiting the spatial diversity due to the movement of the vehicular ends, the framework is able to detect the main scatterers in the environment. In [241], an extension of the GSTM is provided for the discrimination of wrongly estimated targets in mono-static sensing applications. However, the works [217], [241] consider scatterers as “point source” objects leading to individual MPCs. With increasing bandwidths in 5G, and even larger expected ones in 6G, higher sensing resolutions will become available and the geometric extension of objects must be taken into account, as the same authors did in [242].

Whereas the previously mentioned works follow a classical approach based on the definition of MPCs (like traditional communications-centric models) and the relationships between the evolution of their parameters and the geometry and dynamics of the environment, other authors define the ISAC

propagation paradigm as a multi-link scenario comprised of both communication and sensing links [243], [244]. Hybrid models combining multiple technologies are a potential approach. For example, in [245], AI technology is combined with a deterministic method to predict Mobility Profiles (MPs) in a dynamic scenario where vehicles are moving in certain trajectories.

Regardless of the considered approach, a common limitation of the available proposals comes from the fact that most of the models are inherently designed for communications, but rely on some of the communications characteristics to infer information about the environment, rather than providing a model directly related to the physical characteristics of the propagation environment itself. Some proposals try to fill in these gaps by defining channel characteristics directly related to the geometric and kinematic properties of the scatterers (e.g., their sizes, shapes, positions, velocities or movement trajectories), as well as the dynamic nature of the observability of such scatterers (e.g., simultaneous observable number of scatterers or their life-span) [246].

Challenges are still present in relation to ISAC-V channel modeling, such as [126] the need for separating relevant scatterers (i.e., “sensing foreground”) from the “sensing background” (e.g., clutter), as discussed in Section VI-B3. Furthermore, communication channels usually neglect timing and carrier offsets (as they are considered to be compensated by typical receiver architectures), whereas in sensing channels these must be included explicitly as parameters of the channel in order to avoid ranging and speed ambiguities (see Section VI-B2).

C. OPEN CHALLENGES AND OPPORTUNITIES ON CHANNEL MODELING FOR V2X

Despite the field of channel modeling is not new, the upcoming of new scenarios (e.g., NTN) and cases of use (e.g., ISAC-V) in the 6G, along with new opportunities, also brings new challenges. In this subsection, the main identified channel modeling- and propagation-related challenges will be described. Firstly, a short introduction to the general challenges inherent to V2X scenarios will be introduced. Afterwards, the specific challenges in relation to NTN and ISAC-V will be detailed.

1) GENERAL CHALLENGES ASSOCIATED TO V2X SCENARIOS

Considering here the characterization of the propagation of signals in a more aseptic point of view, the challenges posed by V2X in the characterisation of the radio channel model come from the diversity of scenarios and cases that need to be considered. Many of these scenarios involve, in general, the wide mobility of, at least, one of the ends of the radio link, i.e., the mobile terminal. A vehicle can move in a very diverse set of environments, requiring the corresponding propagation models. As a starting point, one can take a survey done for railway communications [251], in which virtually all scenarios have been considered: Viaduct, Cutting,

Tunnel, Urban, Suburban, Rural, Mountain, Water, Platform, In-Carriage, Train-to-Train, Station, Railway Yard, Composite Scenario, UAV-to-Ground, Satellite-to-Ground, Railroad Internet of Things (IoT) and Railway Depot. Despite some of these environments are particular of railways, most of them can be directly used for V2X communications or easily adapted to them, and one can even add a few more: parking places and underground parking (open and in boxes), narrow streets (in old cities, where the width of the street is not much more than the one of the car), traffic jams (where vehicles are very close to each other), different types of vehicles (from small cars to large trucks, and encompassing buses, tractors and many others), among others. These environments pose quite distinct characteristics from the wave propagation viewpoint, and the mere analysis of the existence of LoS is too short to accommodate all required features of a proper model.

2) CHANNEL MODELING AND PROPAGATION CHALLENGES RELATED TO NTN FOR V2X

Although promising, the convergence of terrestrial and NTN link encompasses critical challenges, especially for the case of satellites. Many of these challenges are inherent to the very long propagation distances between the Earth surface and orbiting satellites.

On the one hand, channel non-linearities due to non-ideal Power Amplifiers (PAs) are of utmost relevance [252]. Since satellites are power-limited systems, the PA operation point should be close to the saturation in order to maximize the energy efficiency. However, this operation point is well-known to lead to non-linearities. This challenge can be aggravated by the usage of multicarrier modulations with large PAPR, such as OFDM, although this specific aspect can be partially mitigated with the newest waveform proposals, such as OTFS (see Section VI-B5). OTFS would be beneficial as well to circumvent the large Doppler frequency shifts due to the relative movement between the UEs and the satellites [227], [253], although the usage of OFDM with increased subcarrier spacing is also under consideration [227]. Power control loops become also a challenge due to the very large speeds of the satellites (e.g., up to 27200 km/h for LEO scenarios) and potentially highly-mobile ground vehicles (e.g., high-speed trains). Finally, extra Modulation and Coding Schemes (MCSs) may be required to accommodate very low-SNR schemes.

The large propagation delays also pose great challenges, not only for the propagation channel themselves, but also for different protocol layers, retransmission mechanisms and response times. This is due to the fact that propagation delays can exceed the designed processing times, such as the HARQ processing time [252]. In order to accommodate HARQ strategies to such large delays and to avoid the introduction of a significant jitter, these HARQ strategies may need to be tuned or deactivated in some cases, depending on the round-trip delay. It has been suggested, for example, that the HARQ can be disabled for GEO satellites (round-trip delay of around 542 ms) and the maximum number of

retransmissions reduced for LEO satellites (round-trip delay of tens of milliseconds), depending on the QoS required for the different services [227], [254]. Long propagation delays can also affect to the protocols of initial configuration. Current preambles and synchronization signals can be insufficient to support large round-trip delays or large frequency offsets due to extreme satellite speeds [227].

The combination of terrestrial and aerial network deployments can be challenging as well. The cells served by aerial BSs can cover very large areas and the fast mobility of the aerial and terrestrial nodes can lead to very frequent handovers [252], [254]. Handovers between terrestrial and aerial links can lead to large jitters due to the differences in propagation delay. Besides, these handover challenges affect as well to ISAC-V applications, as mentioned in Section VI-B8. Finally, cell pattern generation strategies are affected as well by large differences in propagation delays between cells.

On the bright side, one may consider that the position of the UE can be assumed as known in most CAV applications, and the trajectory of the satellites can certainly be known in advance. This information can be used at the UE side to adapt the “timing advance” strategies to align uplink and downlink frames as well as to pre-compensate the Doppler frequency effects [227], [252]. Handover strategies can also be planned in advance based on the trajectory of the satellites and the UEs.

A final remark must be done regarding the inhomogeneity and variability of the propagation media. Whereas in purely terrestrial propagation, channel characteristics such as shadowing and multipath fading are essential to characterize the received signals, rain attenuation and scintillation can be more relevant in aerial communications [153], [227]. At the same time, path loss definitions might be updated to take into account the composition of the gases between the aerial transmitter and the terrestrial UE. Clutter and shadow losses are also highly dependent on the elevation angle of the aerial nodes [239].

3) CHANNEL MODELING-RELATED ISAC-V CHALLENGES

Although channel models represent the basic foundation for the optimization of deployments and transceivers (and sensors in ISAC), ISAC-V-enabled channel models are scarce for now. Most of the few available ISAC channel models consider static scenarios. Moreover, even for the channel models considering dynamic situations, most of them can only be applied to the AS sensing mode, neglecting DPS and OPS (see Section VI-A). In order to be applicable to practical applications, ISAC-V channel models should be defined in terms of characteristics directly related to the geometric and kinematic properties of the scatterers (e.g., their sizes, shapes, positions, velocities or movement trajectories), as well as the dynamic nature of the observability of such scatterers (e.g., simultaneous observable number of scatterers or their life-span), rather than just consisting of an interpretable extension of classical communication channels. A final remark on the development of ISAC-V models as mere extensions of traditional channel

models is that the latter usually neglect timing and carrier offsets (as they are considered to be compensated by receiver architectures), which leads to ambiguities in range and target velocity sensing.

VIII. MAIN TAKEAWAYS

Based on the discussion presented throughout this paper, the following conclusions can be drawn regarding the technologies reviewed. First, NTN’s wide-area and high-altitude deployment effectively bridge coverage gaps in terrestrial 6G-V2X networks, enabling continuous communication in remote regions, over oceans, and within airspace. The integrated network architecture supports both Hyper Reliable Low Latency Communication (HRLLC) and URLLC, enabling dual-redundant links for critical scenarios such as autonomous driving and emergency rescue.

As a critical enabler for autonomous driving, real-time cloud/edge service provisioning, high-definition map updates, and vehicle platooning in intelligent transportation ecosystems, URLLC design remains a focal point in V2X ecosystems. However, balancing low latency and high reliability presents inherent challenges, driving considerable research efforts into multiuser access protocols, short blocklength coding/decoding, frame structure optimization, and end-to-end resource slicing. Current techniques have achieved significant milestones, now enabling end-to-end latency below 20 ms with 10^{-5} error rates verified by theory and field tests, satisfying QoS requirements for most V2X applications. While current designs provide best-effort URLLC, realizing truly deterministic performance demands holistic designs, where jitter must be strictly controlled. This challenge stems from the stochastic nature of wireless environment and exponentially expanded data requirement. Evidently, achieving deterministic URLLC performance necessitates cross-layer orchestration spanning from the application layer down to the PHY layer, coupled with the harmonic integration of state-of-the-art techniques.

AI-integration and ISAC have been unequivocally identified as a key components for the future 6G, and in particular for the 6G V2X. AI technology endows 6G-V2X networks with enhanced capabilities in cooperative perception, resource allocation, and trustworthy communication. Despite confronting a range of challenges, the ongoing progress in 6G-AI ensures that the development of 6G-V2X networks will accelerate significantly. These AI-driven networks are destined to be a cornerstone in the eventual realization of a fully ubiquitous intelligent transportation system. For the case of ISAC, it is of paramount importance to highlight that, with the surge of ADAS, autonomous driving and unmanned vehicles, the amount of sensing information generated and required by vehicles is higher than ever. Hence, integrating ISAC into V2X is the natural choice, leading to the so-called ISAC-V. ISAC-V has the potential to (a) alleviate the spectrum congestion by using a common signal for both communication and sensing; (b) allow for the direct CAVs sensor data exchange through the V2X network; (c) improve the communication

services based on the sensing information; and (d) reduce the size, weight and cost of the CAVs on-board hardware. A tortuous path is still to be followed for the consecution of ISAC-V, given the propagation, system design and computational challenges faced.

For any communication system, without V2X being an exception, accurate channel models are an essential tool for the optimization of deployments and the design of transceivers. Typical V2X cases of use are very heterogeneous, leading to a large set of scenarios with very distinctive characteristics and involving relative movement between the communication ends and the surrounding environments. The 3GPP has established a list of service requirements to be fulfilled by V2X and accurate channel models are required in order to realistically evaluate the performance of transceivers and algorithms by vendors. A new scenario and a new case of use, respectively NTN and ISAC, are especially challenging from the propagation and channel modeling point of view. Empirical research, based on on-site radio measurements, as well novel channel modeling paradigms adapted to the new required applications are required to provide the basis support for the new V2X technological enablers.

AUTHORS' DISCLAIMER

Generative artificial intelligence (AI) tools, such as Grammarly and ChatGPT, were employed to improve grammar in certain parts of the paper. The paper's content, core ideas, and references remain the authors' sole responsibility. The use of AI followed ethical guidelines to ensure the originality and accuracy of the work.

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